Probabilistic Seismic Hazard Assessment for Taiwan: Updates and Improvements in TEM PSHA2025

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Highlights

- TEM PSHA2025 is the new version of PSHA for Taiwan.
- Expands coverage to 320 km and integrates multiple international earthquake catalogs.
- Using updated on-land and new identified offshore seismogenic structures.
- Updated ground motion prediction equations and site effects are systematically reassessed.
- Results support seismic mitigation and engineering applications in Taiwan.

Keywords:

Probabilistic seismic hazard assessment (PSHA), Ground motion prediction equations, the Taiwan Earthquake Model (TEM), multiple-structure rupture, offshore seismogenic structures

Abstract

The Taiwan Earthquake Model (TEM) has previously released two probabilistic seismic hazard assessments (PSHA) for Taiwan, TEM PSHA2015 and TEM PSHA2020. Building on these efforts, this study develops TEM PSHA2025, which incorporates substantial updates in earthquake catalog completeness, source characterization, and ground motion prediction equations (GMPEs). The new model expands the study area from only the region surrounding Taiwan to encompass a radius of 320 km, integrates multiple international earthquake catalogs, refines on-land and offshore seismogenic structures with multiple-structure rupture models, and adopts time-dependent recurrence through the Brownian passage time model. In addition, recent Taiwan-specific and global GMPEs are systematically selected and weighted, while site effects have been reassessed using an updated V_{S30} database. Compared to previous versions, TEM PSHA2025 provides a more rigorous treatment of epistemic uncertainties and produces updated hazard maps for Taiwan. These advancements offer a robust scientific foundation for seismic disaster mitigation, regulatory compliance, and engineering applications.

1. Introduction

Advances in Earth sciences have refined PSHA, providing site-specific spectra that improve hazard accuracy and ensure engineering design and risk management remain consistent with current scientific understanding. PSHA has been applied in numerous countries and regions (Mäntyniemi et al., 1984; Mun et al., 2004; Sawires et al., 2005; Liu et al., 2013; Das et al., 2017; Hassan et al., 2020; Peláez et al., 2023). In its most recent version, TEM PSHA2020 (Chan et al., 2020), this assessment incorporates a revised seismogenic structure database (Shyu et al., 2020) with newly identified 3D fault geometries, excluding the Fengshan structure (ID 45). The fault model accounts for both single- and multiple-structure ruptures and adopts the time dependent model, Brownian passage time (BPT) model, for some fault recurrence. An earthquake catalog updated through 2016 is utilized, and advanced shallow-background seismic sources were introduced, with seismicity density evaluated using both areal source models and the smoothing approach of Woo (1996). The model further integrates new ground motion prediction equations (GMPEs), Lin et al. (2011), and incorporates site amplification factors into the hazard calculations.

To support engineering applications and regulatory compliance, and to reflect the assumption of non-crossing seismogenic structures, the shallow areal sources in Taiwan were redefined and expanded by Kao (2024). The TEM fault database has been updated with new geological data on on-land seismogenic structures (Chuang and Shyu, 2025). According to Chang et al. (2023), Coulomb stress analysis was applied to identify on-land seismogenic structures capable of multiple-structure ruptures, and formulas were proposed to calculate the probability of individual ruptures involving two or more structures. Meanwhile, Chuang et al. (submitted) provided geodetic long-term slip rates for selected seismogenic structures, thereby enhancing the interpretation of strike-slip components from a geodetic perspective. For other structures, only geological slip rates were recommended. Additionally, 55 offshore seismogenic structures relevant for assessing the seismic hazard of Taiwan were identified by Chen and Shyu (2025). Their slip rates were estimated using multiple methods, providing robust and reliable values.

This study incorporates a more detailed evaluation of surrounding subduction zones based on the project "Reevaluation of Probabilistic Seismic Hazard of Nuclear Facilities in Taiwan Using SSHAC Level 3 Methodology", hereafter referred to as TWSSHAC Level 3 (National Center for Research on Earthquake Engineering, NCREE, 2015-2019). The subduction interfaces were substantially revised, extending to over 600 km for the Ryukyu zone and ~800 km for the Manila zone, with maximum magnitudes up to M9.0. Each interface is divided into three segments, and multiple-segment ruptures are now considered.

For GMPEs, in addition to the shallow model—Lin (2009) and Lin et al. (2011), which were used in TEM PSHA2015 (Wang et al., 2016) and TEM PSHA2020 (Chan et al., 2020), respectively—and the subduction GMPE of Lin and Lee (2008) applied in both TEM PSHA2015 and TEM PSHA2020, the latest GMPEs have now also been incorporated. These include three recent models developed using local strong-motion datasets (Phung et al., 2020a, 2020b; Chao et al., 2020), as well as the NGA-WEST2 (Boore et al., 2014; Campbell and Bozorgnia, 2014) and NGA-SUB (Abrahamson and Gülerce, 2020) models. The present assessment follows the methodology of Gao et al. (submitted) to provide a more robust and reliable assessment. In this study, all hazard maps considered an epsilon of 2 for the uncertainty of the GMPEs.

Furthermore, the average shear-wave velocity in the upper 30 meters (V_{S30}) map has been updated based on the findings of Chen et al. (2022) and the enhanced TWSSHAC Level 3 site database. This update not only increases the number of borehole measurements but also incorporates microtremor data, providing a more comprehensive and accurate representation of site conditions across Taiwan.

These updates demonstrate a concerted effort to integrate the latest scientific advances and data, ensuring that TEM PSHA2025 delivers a comprehensive and rigorous seismic hazard assessment for Taiwan with important implications for engineering practice. All calculations were performed with the OpenQuake engine (Pagani et al., 2014; Silva et al., 2014), the open-source platform developed by the Global Earthquake Model (GEM) Foundation for seismic hazard and risk analysis using globally recognized datasets and methodologies.

2. Updated the catalog

TEM PSHA2025 includes a comprehensive update of the earthquake catalog, emphasizing improved completeness, precision, and expanded coverage. Unlike TEM

PSHA2015 and TEM PSHA2020, which were limited to Taiwan and nearby offshore regions, the new version extends to a 320 km radius and incorporates events since 1900, in line with the Taiwanese standard (Bureau of Standards, Metrology and Inspection, 2018) "CNS 15176-1:2018 Wind Turbines – Part 1: Design Requirements". Accordingly, this study adopts a broader study area than TEM PSHA2020 and analyzes earthquake data spanning 1900 through the end of 2023.

Within the newly defined study area, local earthquake sources include catalogs from the Central Weather Administration (CWA) Seismic Network (CWASN, 2012), the Broadband Array in Taiwan for Seismology (BATS; Institute of Earth Sciences, Academia Sinica, 1996; Kao et al., 1998; Kao et al., 2000; Chen et al., 2004), and Chang et al. (2016), which documents events with moment magnitude $(M_W) \geq 5.0$ from 1900 to 2014 and was published in a peer-reviewed international journal. Both BATS and Chang et al. (2016) utilize M_W scales.

These datasets provide high-resolution, regionally calibrated seismic data for Taiwan, enhancing earthquake record quality. To address limited local records beyond 119-123°N and 21-26°E, five international catalogs are integrated: the Global Centroid Moment Tensor Catalog (GCMT; Dziewonski et al., 1981; Ekström et al., 2012), United States Geological Survey (USGS; USGS, 2024), National Research Institute for Earth Science and Disaster Resilience (NIED ; NIED, 2024) of Japan, Japan Meteorological Agency (JMA; JMA, 2024) and the International Seismological Centre (ISC; Bondár and Storchak, 2011; Storchak et al., 2017; Storchak et al., 2020; ISC, 2024). This integration mitigates local network limitations.

As CWA records local magnitudes (M_L), and other networks report various magnitude types, M_W is commonly adopted for seismic hazard assessment. The M_L recorded by the CWA are empirically converted to broadband M_W ($M_{w(CWA_{BB})}$), and then to GCMT-compatible M_W ($M_{W(GCMT)}$) using the formula of Cheng et al. (2011):

$$M_L = 0.961 M_{w(CWA_{BB})} + 0.338\ (\pm 0.256)\ (M_L \leq 5.5) \tag{1}$$

$$M_L = 5.115 \ln\left(M_{w(CWA_{BB})}\right) - 3.131\ (\pm 0.379)\ (M_L \geq 5.5) \tag{2}$$

$$M_{W(GCMT)} = 0.951 M_{w(CWA_{BB})} + 0.434\ (\pm 0.14) \tag{3}$$

Surface wave magnitudes (M_S) and body wave magnitudes (mb) are both converted using the empirical formulas proposed by Storchak et al. (2013):

$$M_w = 0.67 M_S + 2.13\ (M_S \leq 6.47) \tag{4}$$

$$M_w = 1.10 M_S - 0.67\ (M_S > 6.47) \tag{5}$$

$$M_w = 1.38 mb - 1.79 \tag{6}$$

The JMA magnitude (M_J), sometimes denoted as M_D or M_V, is converted using the empirical formula established by the Headquarters for Earthquake Research Promotion, Japan (2005, Formula 2-24):

$$M_w = 0.78 M_J + 1.08 \tag{7}$$

To prevent duplicate event representation, a procedure to remove duplicate events was implemented across the multiple earthquake catalogs. Catalogs were ranked by priority (Chang et al., 2016; BATS; CWA; GCMT; USGS; NIED; JMA; ISC), with higher-priority entries retained in cases of overlap. Duplicate events were identified by the following criteria: (1)

temporal separation less than 10 seconds, (2) spatial proximity within 50 km, and (3) M_W difference ≤ 1.0. Catalogs were merged sequentially from lowest to highest priority, and for events meeting all three criteria, only the highest-priority record was preserved. For significant events ($M_W \geq 6.0$) occurring before 1999, when catalog discrepancies were larger, duplicates were further examined if they occurred within the same year, month, day, hour, and tens digit of the minute, with a maximum separation of 3 minutes. In this context, events were defined as duplicates if any two of the three above criteria were met. Following this process, 148,986 unique events ($M_W > 3.0$) were retained from an initial pool of 285,886 entries after removing 136,900 duplicates (Fig. 1).

To ensure suitability for PSHA, the catalog was declustered to satisfy Poisson process assumptions, which require temporal independence of earthquake occurrences. Mainshocks were identified as independent events, while dependent events (foreshocks and aftershocks) were excluded. Unlike TEM PSHA2020, which used only the Gardner and Knopoff (1974) algorithm, this study adopted an enhanced declustering approach (Cheng et al., 2015), combining the algorithms of Gardner and Knopoff (1974), Arabasz and Robinson (1976), Wyss (1979), and Uhrhammer (1986). Events classified as foreshocks or aftershocks under two or more algorithms were removed.

Given spatial and temporal variations in seismic network coverage and catalog completeness, the dataset was divided into three major regions (see Fig. 2. black line): Taiwan island, the China-Taiwan Strait, and the Pacific region (Kao, 2024). Additionally, the completeness of the earthquake catalog in each major region also varies over time due to the evolution of seismic monitoring in Taiwan; therefore, the timeline is divided into four major development stages: 1900–1935, 1936–1972, 1973–1990, and post-1991. These divisions began with low-magnification mechanical seismographs and basic smoked paper recordings, followed by significant advancements after the 1935 Hsinchu-Taichung earthquake; the establishment of the Taiwan Telemetered Seismographic Network (TTSN) in 1973 marked the advent of modern monitoring. Since 1991, the CWA Seismic Network has integrated regional stations, enabling accurate earthquake location throughout Taiwan and its outlying islands.

For the calculation of seismicity using the Gutenberg-Richter relation (Gutenberg and Richter, 1944), Magnitude completeness (M_C) for shallow events was determined for each region and time period provided in Fig. 3 (a-c) by identifying the magnitude bin with the largest event count in the merged raw catalogs. For conservative estimation, M_C thresholds were set slightly above this value. Taiwan's main island, benefiting from denser instrumentation, exhibits lower M_C thresholds in recent years, whereas the China-Taiwan Strait region maintains higher thresholds ($M_C = 6.0$ for 1936–1972) due to sparser records. Furthermore, the deep catalog adopts M_C values of 8.0, 6.0, 5.0, and 4.0 for the four periods (Fig. 3 (d)), respectively.

The final catalog used in this study was compiled by filtering out declustered catalog below the corresponding M_C thresholds. The final shallow catalog will used for shallow areal source analysis, whereas the final deep catalog will be used for subduction intraslab sources.

3. Modification of Shallow Areal Sources

Unlike TEM PSHA2020, which estimated shallow seismicity density using both areal source models and the Woo (1996) smoothing approach, this study exclusively employs areal sources. This choice reflects the absence of clear evidence for the superior accuracy of smoothing methods and ensures more efficient computational use. A key enhancement in TEM PSHA2025 shallow areal sources is the adoption a broader study area consistent with the catalogs, with source depths restricted to shallower than 35 km.

Regional divisions (Kao, 2024) integrate seismicity, geological structures, geophysical data, and previous studies. Notably, the S07 and S08 regions have been merged into S07 to maintain geological continuity, as updated fault lines (Chuang and Shyu, 2025) intersect the source regions defined by Kao (2024); all regions numbered S09 and above in Kao (2024) have now been renumbered.

The truncated exponential model (also known as the truncated Gutenberg-Richter relation or TED) is a modification of the Gutenberg-Richter relation (Gutenberg and Richter, 1944), in which both a minimum and an upper magnitude (M_u) are specified. Seismicity parameters are estimated using the maximum likelihood estimation (MLE) method to determine the *b*-value of the Gutenberg-Richter relation for the entire study area (Fig. 2. top left), with subregional *b*-values derived from this overall value. MLE provides more robust statistical estimates than least squares approach. The 2022 Chishang earthquake (M_W7.0), which did not involve complete rupture of any mapped fault and therefore does not conform to the characteristic earthquake model used in this study, it is classified as a background event, leading to the upper magnitude (M_u) for Taiwan's main island being adjusted from 6.9 to 7.0. For offshore regions, M_u due to higher uncertainties and the absence of events matching existing seismotectonic models, the M_u thresholds follow TWSSHAC Level 3: maximum observed magnitude (1900–2023) plus 0.25 (Fig. 2). In our model, all the minimum magnitudes of truncated exponential model are set a 4.0.

Updated shallow areal sources result in revised seismic hazard estimates (Fig. 4 (a)). Compared to TEM PSHA2020 (Fig. 4 (b)), central Taiwan—especially the S10 region—shows a significant hazard decrease, while a slight increase is observed in Taipei. These changes are attributed to differences in catalog time spans and updated regional source delineation.

4. Modification of On-Land Seismogenic Structure

4.1. Updated seismogenic structure database

The update of on-land seismogenic structures constitutes a significant advancement in source characterization for TEM PSHA2025. The seismogenic parameter table (Table 1) now includes 48 faults, revised from 44 in TEM PSHA2020, based on Chuang and Shyu (2025), with corresponding parameter updates. Major revisions include redefining the Chungchou structure (ID 23) into the North and South Chungchou structures (IDs 23 and 46), and the addition of the Kouhsiaoli fault (ID 47) and Chekualin fault (ID 48). The types of updated fault modifications are distinguished in Fig. 5 by color: dark-violet for faults with modified geometry and parameters; seafoam green for faults with parameter adjustments; red for newly added faults; gray for unchanged faults; and blue for new offshore seismogenic structures addressed in a later section. This update reflects recent progress in Taiwan's active tectonic research and ongoing efforts to improve seismic hazard assessments.

The multiple-structure rupture model, as applied in TEM PSHA2020 and Chang et al. (2023), assesses the potential for coseismic ruptures resulting in larger earthquakes. Pairings for possible multiple-structure ruptures (Table 2) have been updated using Coulomb stress calculations from Chang et al. (2023), providing a more physically objective pairing method versus expert judgment. The pairing of the Shihtan fault (ID 13) and Tuntzuchiao fault (ID 15) is retained to reflect the 1935 Hsinchu-Taichung earthquake, which is attributed to a coseismic rupture involving these two faults, despite its exclusion from Chang et al. (2023). Additionally, due to substantial changes in the Chungchou fault (ID 23) geometry by Chuang and Shyu (2025), all associated multiple-structure pairings from Chang et al. (2023) have been removed to maintain model consistency.

The application of the BPT model has been expanded (see Table 3; values derived from M_W (Y&M) and mean slip rate), notably including the Shanchiao fault, thereby providing a more accurate representation of earthquake recurrence and temporal dependence. The BPT calculation procedure is consistent with that used in TEM PSHA2020, but rupture probabilities for the next 50 years are now calculated starting from 2025.

The uncertainties in relevant parameters have a significant impact on seismic hazard results, and accounting for model uncertainty enhances the credibility of seismic hazard assessments (Bommer, 2003; Wang et al., 2012). Uncertainty quantification is enhanced by considering both M_W(W&C) and M_W(Y&M) with equal weights. M_W(W&C) uncertainty is incorporated via ± 1 standard deviation (σ) with weights of 0.6 (mean) and 0.2 (bounds), following Wells and Coppersmith (1994); M_W(Y&M) uses a single value due to lack of associated σ (Yen and Ma, 2011). Slip rate uncertainty is addressed per Chuang and Shyu (2025), with maximum and minimum values reflecting extreme, not statistically distributed, conditions; a weighting scheme of 0.05-0.90-0.05 is used for maximum, mean, and minimum values, respectively. The resulting logic tree (Fig. 6) more accurately reflects epistemic uncertainty in seismic hazard characterization.

Seismic hazard results following the updated on-land structure database are shown in Fig. 7 (a), with differences from the prior model (Fig. 4 (a)) in Fig. 7 (b). Observed changes primarily derive from slip rate modifications, explicit consideration of uncertainties, adjustments in Chungchou fault geometry, and revised multiple-structure rupture pairings, among other factors. For example, weighted average slip rates that higher than previous values for the Hukou fault (ID 4), Touhuanping structure (ID 9), and Miaoli frontal structure (ID 10) explain elevated hazards in Taoyuan, Hsinchu, and Miaoli. Similarly, higher weighted average slip rates for the Hsiaokangshan fault (ID 27) and Hengchun offshore structure (ID 31) raise hazard in their respective regions, while a lower weighted average slip rate for the Chushiang structure (ID 39) decreases local hazard. Geometry changes to the Chungchou structure led to reduced hazards near its former location. The addition of new structures (IDs 45–48) increased hazard in their surrounding areas. Updates to multiple-structure rupture pairings modified the relative probabilities of single and multiple ruptures, and use of "-1" fault data in some cases altered fault areas, impacting M_W, recurrence intervals, and regional seismic hazard.

4.2. Geodetic Slip Rate

The slip rates used in the previous section were based on geologically derived rates from field investigations. In this update, geologic slip rates for 21 on-land seismogenic structures have also been incorporated (Chuang et al., submitted). This integration of multiple data sources enables a more comprehensive and reliable assessment of fault activity.

To obtain a reasonable geodetic slip rate for seismic hazard assessment, we adopt the slip deficit rate inferred from geodetic data. Since different models and data spatial resolutions may yield inconsistent slip deficit rates, we determined a representative geodetic slip rate for each fault through an expert panel discussion, based on the results of two modeling approaches: Baseline Inversion and tDEFNODE (McCaffrey, 2009). The selection was guided by the following criteria:

For cases where the slip deficit rates from both models are generally consistent with the geological slip rate, typically with a difference of less than 3 mm/yr, we adopt the average of the two modeled slip deficit rates.

For cases where the slip deficit rates from both models are mutually consistent but differ from the geological slip rate, and the fault is confirmed to have a significant strike-slip component, we compare the dip-slip component of the geodetic slip deficit rate against the geological slip rate. If the dip-slip component and geological slip rate agree, we adopt the average of the two modeled slip deficit rates.

For cases in which the slip deficit rates from the two models are inconsistent, but one model yields a slip deficit rate that closely matches the geological slip rate, we adopt the modeled slip deficit rate that agrees with the geological rate.

For cases where the slip deficit rates from the two models are inconsistent, and the fault exhibits a significant strike-slip component, but the dip-slip component of one modeled slip deficit rate is consistent with the geological slip rate, we adopt the modeled slip deficit rate whose dip-slip component matches the geological slip rate.

For the 21 structures with both geological and geodetic slip rate data (Table 1), this study integrates the two sources by assigning each a probability weighting of 0.5 (Fig. 6). All other uncertainties are treated as in the previous section. For the remaining structures, only the geologic slip rate is used.

After incorporating the geologic slip rates for 21 faults, the resulting seismic hazard is presented in Fig. 8 (a), while the differences compared to the previous one (Fig. 7 (a)) are shown in Fig. 8 (b). Except for the Changhua fault (ID 16) and Chelungpu fault (ID 17), for which changes are not apparent due to application of the BPT model, differences in the hazard map largely correspond to variations in slip rates. For example, the hazard surrounding the Chaochou fault (ID 29) increased by approximately 0.1 g due to higher geodetic slip rates. In contrast, decreases are observed around the Central Range fault (ID 34) by 0.1 g, the Kouhsiaoli fault (ID 47) by 0.1–0.2 g, and the Fengshan River strike-slip structure (ID 5) by 0.05 g, all resulting from geodetic slip rates that are lower than geologic slip rates.

5. Integration of the Offshore Seismogenic Structure Database

The incorporation of offshore seismogenic structures represents a major advancement in the TEM PSHA2025, enabling a more comprehensive evaluation of the contributions from seismogenic structure source. This study adopts the offshore structure database (Table 4) developed by Chen and Shyu (2025), which includes several proposed multiple-structure rupture scenarios, including 01+O01, 06+O52, 08+O53, and O54A+O54B (Table 2). In addition, slip rates reported by Chen and Shyu (2024) are used to estimate the recurrence intervals of individual structures.

Based on the interpretation of seismic reflection profiles and bathymetric data, Chen and Shyu (2025) identified 55 offshore structures near Taiwan that are capable of generating earthquakes with magnitudes of $M \geq 6.5$. The subsurface geometries of these structures were reconstructed using regional seismogenic depth estimates and fault cross-cutting relationships. Structural activity was assessed by integrating geological and geophysical datasets, and both slip rates and recurrence intervals were estimated using regional tectonic models (Chen and

Shyu, 2024). All uncertainties for offshore seismogenic structures are treated in the same manner as for on-land seismogenic structures (Fig. 6), as described in the previous section, except that geodetic slip rates are not included.

The results show an increased seismic hazard (Fig. 9) along the coastal areas of Hsinchu and Kaohsiung due to the proximity of multiple offshore structures. In contrast, the northeastern offshore region, which is dominated by normal faults with longer recurrence intervals, exerts only a limited impact on the seismic hazard under the 475-year return period scenario.

6. Subduction Zone Source Model

6.1. Subduction Interface

Taiwan's subduction zone system, formed by the convergence of the Philippine Sea Plate and the Eurasian Plate, produces significant seismicity and complex tectonic interactions. Accurate geometric characterization of subduction zone interface earthquakes is essential, as these events often have larger magnitudes and extensive impact. The updated subduction interface model utilizes TWSSHAC Level 3 interface geometry and average dip angles from multiple sources (Lin et al., 2009; Theunissen et al., 2010; Klingelhoefer et al., 2012; Lallemand et al., 2013).

Major subduction interface earthquakes frequently involve rupture across multiple segments, exemplified by the 2011 Tohoku and 2004 Sumatra earthquakes. To capture this behavior, our study allows for both adjacent and entire segment ruptures in each subduction zone interface, applying the previously mentioned multiple-structure rupture model for seismogenic structures (Chang et al., 2023).

The M_W of subduction interface earthquakes is estimated using rupture area (A, in km^2) or rupture length (L, in km), following the scaling laws of Strasser et al. (2010) and Blaser et al. (2010). The formulas for $M_w\_Strasser(A)$ and $M_w\_Strasser(L)$ based on rupture area and length, as derived from Strasser et al. (2010), are as follows:

$$M_w\_Strasser(A) = \log(Area) \times 0.846 + 4.441, \sigma = 0.286 \quad (8)$$

$$M_w\_Strasser(L) = \log(Length) \times 1.392 + 4.868, \sigma = 0.277 \quad (9)$$

The formula for $M_w\_Blaser(L)$ based on rupture length, as derived from Blaser et al. (2010) and can be represented as follows:

$$M_w\_Blaser(L) = (\log(Length) + 2.81) \div 0.62, \sigma = 0.16 \quad (10)$$

Three M_W estimates are considered, with weights of 0.4, 0.3, and 0.3, as specified by TWSSHAC Level 3. For each M_W, the mean is weighted at 0.6, and the lower and upper bounds ($\pm 1\sigma$) at 0.2 each, consistent with TWSSHAC Level 3 methodology.

Slip rates for the subduction interfaces also follow TWSSHAC Level 3 adopted values: Ryukyu rates of 30, 25, and 15 mm/yr, and Manila rates of 24, 14, and 8 mm/yr. Uncertainty is addressed by considering all three rates, with the mean weighted at 0.4 and the other values at 0.3 each. Uncertainties in subduction interface parameters, including M_W and slip rate, are incorporated through a logic tree (Fig. 6), representing epistemic uncertainty comprehensively. Table 5 presents the mean $M_W\_Strasser(A)$ and mean slip rate, along with the corresponding calculated recurrence intervals as examples.

6.2. Subduction Intraslab

Intraslab earthquake modeling for subduction zones utilizes the truncated exponential model, which effectively represents seismicity. Regional divisions of intraslab sources for the Ryukyu and Manila subduction zones are based on the TWSSHAC Level 3 three-dimensional plate model and earthquake records from 1900 to 2023. The Ryukyu zone (35–300 km depth) is subdivided into five regions (NP1–NP5), while the Manila zone (40–220 km depth) is divided into four regions (SP1–SP4); depth boundaries for each subregion are defined in Table 6.

Seismicity rates for each subduction intraslab source are determined using the truncated exponential model, applied to depth-specific portions of the deep earthquake catalog, with stable b-values derived from local seismicity. For both the Ryukyu and Manila zones, a single b-value per subduction zone is adopted based on the MLE method to ensure consistency; corresponding a-values are then calculated for each region. Estimated a- and b-values for all nine intraslab regions are shown in Fig. 10 (a).

Seismic hazard results after updating the subduction model are shown in Fig. 10 (a). For the 475-year PGA, hazard changes to the previous one are limited to within ±0.025 g, indicating minimal variation. For the 475-year SA1.0 (long-period), slight increases in hazard across the western plains of Taiwan (Fig. 10 (c)); for the 2475-year SA1.0, larger increases are observed, especially in alluvial and basin regions such as Pingtung and Ilan Plains and the Taipei Basin

(Fig. 10 (d)).

7. Selection and Weighting of Ground Motion Prediction Equations

For shallow crustal sources, the study will implement GMPEs developed for the Taiwan region, include the models by Chao et al. (2020), Lin (2009), Lin et al. (2011), and Phung et al. (2020a). In addition, to address model uncertainty and incorporate internationally recognized models, the Next Generation Attenuation Relationships for Western US (NGA-West2, https://peer.berkeley.edu/research/nga-west-2) GMPEs—based on global datasets—will be included. These consist of Abrahamson et al. (2014), Boore et al. (2014), Campbell and Bozorgnia (2014), Chiou and Youngs (2014), and Idriss (2014).

For subduction zone sources, encompassing both interface and intraslab earthquakes, the study will apply GMPEs appropriate for Taiwan. These include Chao et al. (2020), Lin and Lee (2008), and Phung et al. (2020b). To ensure comprehensive treatment of epistemic uncertainty and broaden applicability, NGA-Sub GMPEs derived from global datasets will also be incorporated. These models include Abrahamson and Gülerce (2020), Kuehn et al. (2020), Si et al. (2022), and Parker et al. (2022).

Following the methodology of Gao et al. (submitted), weights are assigned according to the comprehensive selection procedure (SP method) proposed by Salic et al. (2017), which integrates the strengths of multiple evaluation metrics, including Loglikelihood (LLH; Scherbaum et al., 2009) and Euclidean Distance Based Ranking (EDR; Kale and Akkar, 2013).

After a comprehensive evaluation, the following weights were given: For shallow crustal sources, which included shallow areal sources and seismogenic structures, Chao et al. (2020), Lin (2009), Lin et al. (2011), Phung et al. (2020a), Boore et al. (2014), and Campbell and Bozorgnia (2014) received weights of 0.183, 0.156, 0.200, 0.169, 0.146, and 0.146, respectively. For subduction zone sources, Chao et al. (2020), Lin and Lee (2008), Phung et al. (2020b), and Abrahamson and Gülerce (2020) received weights of 0.232, 0.256, 0.276, and 0.236, respectively. This result underscores the superior performance of regionally calibrated GMPEs, and the findings are consistent with Gao et al. (2021).

The hazard map using the new combination of GMPEs is shown in Fig. 11 (a). Due to differences in magnitude and distance scaling among the various GMPEs, the results from using multiple GMPEs differ from those obtained with the single GMPE (Lin et al., 2011) used in TEM PSHA2020, as illustrated in (Fig. 11 (b)). These differences represent the cumulative effects of all the GMPE variations.

8. Integration and Update of V_{S30} Maps

The updated V_{S30} map integrates findings from Chen et al. (2022) and the TWSSHAC Level 3 site database (Kuo et al., 2017), and incorporates significant results from key studies focused on Taiwan's shallow shear-wave velocity structures and site classifications (Kuo et al., 2009; Lin et al., 2009; Kuo et al., 2011; Kuo et al., 2012; Kuo et al., 2016; Lin et al., 2018). Both the V_{S30} and shear wave velocity to 1.0 km/s (Z1.0) maps are generated using a two-stage interpolation process: initial inverse distance weighting (IDW) followed by refinement through linear interpolation, resulting in a final V_{S30} map with final 2-km grid resolution.

V_{S30} plays a crucial role in ground motion prediction, especially across the extensive western plains of Taiwan where soft sedimentary layers are prevalent. The new V_{S30} maps fully leverage geophysical investigations conducted throughout Taiwan, including site surveys at seismic stations, geotechnical borehole records, and geophysical exploration results. Integrating these diverse data sources ensures the reliability and representativeness of the values. The differences between the new and previous V_{S30} maps are shown in the Fig. 12 (a).

However, some GMPEs require specific site conditions such as Z1.0 and the depth to a shear wave velocity of 2.5 km/s (Z2.5), reflecting an enhanced understanding of sedimentary basin effects. Z1.0 is derived from Taiwan's shallow velocity structure (Kuo et al., 2016):

$$\ln(Z_{1.0}) = \frac{-3.8}{2}\ln\left(\frac{V_{S30}^2 + 266^2}{1750^2 + 266^2}\right) \tag{11}$$

Z2.5 is determined using the conversion equations proposed by Kaklamanos et al. (2011), which provide methods for estimating unknown input values when implementing the NGA ground-motion prediction equations. The following empirical relations were used:

$$Z_{2.5} = 519 + 3.595 \times Z_{1.0} \tag{12}$$

After updating the V_{S30} map, the PGA differences compared to the previous one (Fig. 11 (a)) is shown in Fig. 12 (b), while the SA1.0 differences is shown in Fig. 12 (c). The differences in SA1.0 are much more pronounced than those in PGA.

9. Discussion and Conclusions

The development of TEM PSHA2025 marks a significant advancement in seismic hazard assessment for Taiwan. Through expanding and refining earthquake catalogs, improving source models, enhancing site effect evaluation, and optimizing the selection of GMPEs, the new version achieves notable improvements across multiple facets. The final results include a mean hazard map (Fig. 13 (a)) that incorporates site amplification effects, representing a 10% probability of exceedance in 50 years for PGA (in g), intended to address public information needs. Although this study presents the mean hazard map, some studies suggest that the median offers greater stability (Valentini and Beltran, 2024), as the mean is susceptible to extreme values and can result in higher estimated seismic hazard than the median. As a result, the use of the mean has been challenged (Abrahamson and Bommer, 2005)—an effect that becomes more pronounced with longer return periods.

Therefore, Fig. 13 (b) shows the median hazard map within the complete TEM PSHA framework, and Fig. 13 (c) illustrates the difference between the median and mean values (median minus mean). The most pronounced decrease in the median hazard map occurs in eastern Taiwan, primarily due to the Longitudinal Valley fault, whose maximum slip rate (extreme value) is 18.68 mm/yr, while its mean value is only 11.35 mm/yr. Because the mean hazard map is influenced by extreme values, it indicates higher hazards, resulting in lower hazards in the median hazard map.

The results also include the map incorporating for site amplification effects at a 2% probability in 50 years for PGA (in g) (Fig. 13 (d)), as well as the median hazard map for engineering bedrock (V_{S30} = 760 m/s) at 10% probability in 50 years, which is presented in the Appendix to support engineering applications (Fig. a).

For future directions, further research is encouraged to reduce uncertainties and improve the accuracy of PSHA. First, methods for merging earthquake catalogs should be refined to ensure more comprehensive data consistency. Continued investigation into fault rupture histories will benefit the development of time-dependent models such as the BPT model. The optimal selection of upper magnitude limits for characteristic earthquake models also warrants careful evaluation, as this directly influences hazard estimates. Clarifying the source fault for significant events such as the April 3, 2024 earthquake (M_W7.3), which may have originated from an offshore seismogenic structure, such as the Takangkou High thrust (ID 22) or the Chimei Canyon thrust (ID 23), as suggested by Chen and Shyu (2025), remains an ongoing challenge.

Additionally, selecting between purely characteristic earthquake models (which assume entire rupture of a fault, as in our model) and generalized characteristic models (which allow for earthquakes of various magnitudes along a fault) is critical for Taiwan's context. It is especially important to choose a model that minimizes the risk of double-count hazard when combined with truncated exponential models for shallow areal sources. Updated GMPEs (e.g., Phung et al., 2023) may be incorporated. Future efforts could consider using or comparing the branch weighting mechanism proposed by Wang and Lin (2021) with the current approach for assigning branch weights in the GMPE logic tree.

Finally, the adoption of multiple-modal and logic tree approaches can further enhance the representation of epistemic uncertainties in seismic hazard studies. With advancements in monitoring technology and scientific understanding, the TEM model will continue to be updated, providing an increasingly robust scientific foundation for earthquake disaster mitigation in Taiwan. Given Taiwan's complex geological setting and dense population, accurate seismic hazard assessment is crucial for mitigating earthquake risks, informing government disaster prevention policies, establishing engineering safety standards, and enabling sound risk evaluation for the insurance industry. Continued methodological improvements and scientific research will ensure that TEM PSHA remains central to seismic risk management and disaster preparedness in Taiwan.

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Data availability

The earthquake catalog, seismogenic structure parameters, and site condition datasets used in TEM PSHA2025 are available from open and institutional sources, including the Central Weather Administration Seismic Network (CWA), Broadband Array in Taiwan for Seismology (BATS), GCMT, USGS, NIED, JMA, and ISC. The updated fault database follows Chuang and Shyu (2025) and Chen and Shyu (2025). Processed datasets and hazard calculation results generated in this study are available upon reasonable request to the corresponding author.

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Table 1. Parameters for on-land seismogenic structures.

ID	*Seismogenic structure name*	*Fault type* [#1]	*Length (km)*	*Widt h (km)*	*Area (km^2)*	M_w*(W&C)*	M_w*(Y&M)*	*Slip rate (mm/yr)*	*Recurrence interval (yr)* [#2]
1-1	Shanchiao fault	N	54.1	19.44	1051.7	7.01 (±0.25)	7.02	1.66 (1.1–2.94) 1.56 (0.42-2.7) [#3]	807
2	Shuanglienpo structure	R	18.7	11.97	223.84	6.44 (±0.25)	6.35	0.13 (0.06–0.52)	4762
3	Yangmei structure	R	22.1	3.46	76.47	6.03 (±0.25)	5.88	0.21 (0.11–0.74)	1715
4	Hukou fault	R	25.6	20	512	6.77 (±0.25)	6.71	0.8 (0.36–3.65)	1175
5-1	Fengshan River strike-slip structure	SS	30.6	20.08	614.45	6.82 (±0.23)	6.79	3.17 (1.8–10) 1.61 (0.37–2.85) [#3]	325
6	Hsinchu fault	R	14.5	14.14	205.03	6.41 (±0.25)	6.31	0.4 (0.18–1.57)	1475

ID	Seismogenic structure name	Fault type [#1]	Length (km)	Width (km)	Area (km^2)	M_w(W&C)	M_w(Y&M)	Slip rate (mm/yr)	Recurrence interval (yr) [#2]
7-1	Hsincheng fault	R	28.5	37.5	1068.75	7.06 (±0.25)	7.03	0.88 (0.39–4.09)	1546
8	Hsinchu frontal structure	R	12.1	20	242	6.48 (±0.25)	6.38	1.38 (0.59–6.64)	464
9-1	Touhuanping structure	SS	25.8	12.55	323.79	6.54 (±0.23)	6.51	1.95 (0.71–3.37)	385
10	Miaoli frontal structure	R	30.9	15.16	468.44	6.73 (±0.25)	6.67	2.94 (2.58–3.3)	306
11	Tunglo structure	R	15.7	7	109.9	6.17 (±0.25)	6.04	0.5 (0.19–2.63)	860
12	East Miaoli structure	R	14.4	8	115.2	6.19 (±0.25)	6.06	0.84 (0.36–3.89)	524
13-1	Shihtan fault	R	30.7	20.71	635.8	6.85 (±0.23)	6.8	1.38 (0.61–5.32)	746
14	Sanyi fault	R	29.8	34.77	1036.15	7.04 (±0.25)	7.02	0.97 (0.41–4.61)	1403
15-1	Tuntzuchiao fault	SS	27	25.1	677.7	6.87 (±0.23)	6.83	0.47 (0.23–1.67)	2299

ID	Seismogenic structure name	Fault type [#1]	Length (km)	Width h (km)	Area (km^2)	M_w(W&C)	M_w(Y&M)	Slip rate (mm/yr)	Recurrence interval (yr) [#2]
16	Changhua fault	R	82.2	48.55	3990.81	7.57 (±0.25)	7.77	1.77 (0.9–6.63) 3.48 (0.97–5.99) [#3]	2653
17	Chelungpu fault	R	91.9	46.36	4260.48	7.60 (±0.25)	7.81	6.94 (6.94–6.94) 2.72 (0.78–4.65) [#3]	728
18	Tamaopu - Shuangtung fault	R	69.2	12	830.4	6.96 (±0.25)	6.92	1 (0.42–4.88)	1200
19-1	Chiuchiungkeng fault	R	33.6	20	672	6.87 (±0.25)	6.83	4.66 (1.87–23.39) 3.2 (0.1–7.4) [#3]	234
20-1	Meishan fault	SS	25.2	10.04	253.01	6.43 (±0.23)	6.4	2.51 (2.5–2.54) 1.35 (0.3–2.41) [#3]	259

ID	Seismogenic structure name	Fault type #1	Length (km)	Width (km)	Area (km^2)	M_w(W&C)	M_w(Y&M)	Slip rate (mm/yr)	Recurrence interval (yr) #2
21	Chiayi frontal structure	R	34.1	46.36	1580.88	7.21 (±0.25)	7.24	3.36 (1.4–16.12)	566
22	Muchiliao - Liuchia fault	R	26.4	24	633.6	6.85 (±0.25)	6.8	5.75 (4.4–7.1)	181
23	North Chungchou structure	R	18.6	7.45	138.57	6.26 (±0.25)	6.14	3.43 (1.3–6.28)	143
24-1	Hsinhua fault	SS	14.8	11.29	167.09	6.25 (±0.23)	6.22	2.65 (0.8–4.5) 2.2 (0.1–5) #3	200
25	Houchiali fault	R	12.2	5.42	66.12	5.97 (±0.25)	5.82	6.2 (2.65–9.75) 6.3 (1.9–10.7) #3	55
26-1	Chishan fault	SS/R	40	13.82	552.8	6.78 (±0.23)	6.74	1.1 (0.72–1.5)	882
27	Hsiaokangshan fault	R	10.6	12.21	129.43	6.23 (±0.25)	6.11	5.18 (4.26–6.61)	91

ID	*Seismogenic structure name*	*Fault type* [#1]	*Length (km)*	*Width h (km)*	*Area (km^2)*	M_w*(W&C)*	M_w*(Y&M)*	*Slip rate (mm/yr)*	*Recurrence interval (yr)* [#2]
								3.2 (0.8–5.6) [#3]	
28-1	Kaoping River structure	SS/R	33.4	20.71	691.71	6.88 (±0.23)	6.84	0.22 (0.11–0.69)	4950
29-1	Chaochou fault	SS/R	99.3	16.82	1670.23	7.27 (±0.23)	7.27	0.7 (0.4–2.26) 4.2 (1.21–7.18) [#3]	2857
30-1	Hengchun fault	SS/R	42.4	25.88	1097.31	7.08 (±0.23)	7.04	4.45 (2.89–6.14)	308
31	Hengchun offshore structure	R	20.4	10	204	6.41 (±0.25)	6.31	3.84 (1.74–7.95)	154
32	Milun fault	SS/R	32.6	10.35	337.41	6.56 (±0.23)	6.53	10.15 (9.92–10.47)	76
33	Longitudinal Valley fault	R/SS	147.5	23.79	3509.02	7.52 (±0.25)	7.7	11.35 (5.13–18.68)	369
34	Central Range structure	R	86.2	28.28	2437.74	7.38 (±0.25)	7.49	7.28 (4.76–11.16)	401

ID	Seismogenic structure name	Fault type [#1]	Length (km)	Width (km)	Area (km^2)	M_w(W&C)	M_w(Y&M)	Slip rate (mm/yr)	Recurrence interval (yr) [#2]
								3.16 (0.57–5.75) [#3]	
35	Luyeh fault	R	19.6	6.83	133.87	6.24 (±0.25)	6.13	5.28 (3.55–8.02) 3.26 (0.9–5.61) [#3]	93
36-1	Taimali coastline structure	R/SS	43	15.53	667.79	6.87 (±0.25)	6.82	7.32 (5.74–9.03)	143
37-1	Northern Ilan structure	N	74.9	24.54	1838.05	7.26 (±0.25)	7.32	3.29 (0.96–6.27) 1.7 (0.7–2.7) [#3]	657
38-1	Southern Ilan structure	N	21.9	18.04	395.08	6.58 (±0.25)	6.6	5.48 (4.47–6.92) 5.5 (0.1–10.9) [#3]	151
39	Chushiang structure	R/SS	19.8	3.66	72.47	6.00 (±0.25)	5.86	2.9 (1.3–4.5)	121

ID	Seismogenic structure name	Fault type [#1]	Length (km)	Width h (km)	Area (km^2)	M_w(W&C)	M_w(Y&M)	Slip rate (mm/yr)	Recurrence interval (yr) [#2]
40	Gukeng structure	SS	9.2	12.05	110.86	6.07 (±0.23)	6.04	0.94 (0.56–2.52) 2.3 (0.1–6.7) [#3]	457
41	Tainan frontal structure	R	32.9	40.77	1341.33	7.14 (±0.25)	7.14	0.92 (0.45–3.5) 3.24 (0.91–5.56) [#3]	1727
42	Longchuan structure	R	20.7	13.86	286.35	6.54 (±0.25)	6.46	1.7 (0.6–7.83) 0.3 (0.1–1) [#3]	418
43	Youchang structure	R/SS	15.5	12.42	192.51	6.39 (±0.25)	6.28	1.64 (0.92–5.46)	348
44-1	Fengshan hills frontal structure	R	19.1	35	668.5	6.87 (±0.25)	6.83	0.92 (0.4–4.24)	1185
45-1	Fengshan structure	SS/R	16.8	17.57	295.18	6.50 (±0.23)	6.47	3 (1–5)	237

ID	Seismogenic structure name	Fault type #1	Length (km)	Width (km)	Area (km^2)	M_w(W&C)	M_w(Y&M)	Slip rate (mm/yr)	Recurrence interval (yr) #2
								2 (0.1–5) #3	
46	South Chungchou structure	R	14.6	7.45	108.77	6.16 (±0.25)	6.04	2.84 (0.62–5.8) 3.4 (0.6–6.2) #3	155
47	Kouhsiaoli fault	R	20.9	7.47	156.12	6.30 (±0.25)	6.19	9.85 (3.01–20.38) 1.4 (0.1–11.3) #3	52
48-1	Chekualin fault	R/SS	25.6	17.32	443.39	6.71 (±0.25)	6.65	5.55 (2.71–9.23) 3.3 (0.1–8.9) #3	159

Note: #1: R: reverse fault; SS: strike-slip fault; N: normal fault; SS/R: strike-slip dominated fault with minor reverse motion; R/SS: reverse dominated fault with minor strike-slip motion.

#2*: Used M_w(Y&M) and mean geologic slip rate*

#3*: Geodetic slip rate*

Table 2. The pairing and associated parameters for multiple-structure ruptures, where structures with IDs O52~O54B represent offshore seismogenic structure.

ID [#1]	***Seismogenic structure name***	***Type*** [#2]	***Area (km^2)***	***M_w(W&C)***	***M_w(Y&M)***	***Recurrence interval (yr)*** [#3]
01+O01	Shanchiao fault, Outer Chinshan structure 1(ST-Ⅰ-2)	N, N	1312.875	7.11 (6.86–7.36)	7.13	1907
02+03	Shuanglienpo structure, Yangmei structure	R, R	300.31	6.56 (6.31–6.81)	6.48	11332
02+04	Shuanglienpo structure, Hukou fault	R, R	735.84	6.91 (6.66–7.16)	6.87	7253
04+05 [#4]	Hukou fault, Fengshan River strike slip structure	R, SS	1126.45	7.09 (6.86–7.32)	7.05	1264
04+06	Hukou fault, Hsinchu fault	R, R	717.03	6.9 (6.65–7.15)	6.86	6610
06+08	Hsinchu fault, Hsinchu frontal structure	R, R	447.03	6.72 (6.47–6.97)	6.65	2820
06+09	Hsinchu fault, Touhuanping structure	R, SS	528.82	6.76 (6.53–6.99)	6.72	2175
06+O52	Hsinchu fault, Outer Hsinchu structure	R, R	395.9488	6.67 (6.42–6.92)	6.6	5419

ID [#1]	*Seismogenic structure name*	*Type* [#2]	*Area (km^2)*	*M_w(W&C)*	*M_w(Y&M)*	*Recurrence interval (yr)* [#3]
08+O53	Hsinchu frontal structure, Toufen structure	R, R	429.9937	6.7 (6.45–6.95)	6.63	1343
09+10	Touhuanping structure, Miaoli frontal structure	SS, R	792.234	6.94 (6.69–7.19)	6.9	1288
10+15	Miaoli frontal structure, Tuntzuchiao fault	R, SS	1146.144	7.1 (6.87–7.33)	7.06	2422
11+14	Tunglo structure, Sanyi fault	R, R	1146.05	7.08 (6.83–7.33)	7.06	4556
13+14	Shihtan fault, Sanyi fault	R, R	1671.95	7.23 (6.98–7.48)	7.27	4407
13+15	Shihtan fault, Tuntzuchiao fault	R, SS	1313.5	7.16 (6.93–7.39)	7.13	4787
19[#4]+22	Chiuchiungkeng fault, Muchiliao-Liuchia fault	R, R	1305.6	7.13 (6.88–7.38)	7.13	510
20[#4]+21	Meishan fault, Chiayi frontal structure	SS, R	1833.89	7.27 (7.02–7.52)	7.32	1902
21+41[#4]	Chiayi frontal structure, Tainan frontal structure	R, R	2922.21	7.45 (7.2–7.7)	7.59	3413

ID [#1]	*Seismogenic structure name*	*Type* [#2]	*Area (km²)*	M_w*(W&C)*	M_w*(Y&M)*	*Recurrence interval (yr)* [#3]
23+46[#4]	North Chungchou structure, South Chungchou structure	R, R	247.34	6.48 (6.23–6.73)	6.39	356
24[#4]*+25*[#4]	Hsinhua fault, Houchiali fault	SS, R	233.21	6.4 (6.17–6.63)	6.37	293
26+45[#4]	Chishan fault, Fengshan structure	SS/R, SS/R	847.98	6.97 (6.74–7.2)	6.93	1444
33a	Longitudinal Valley fault (North)	R/SS	913.54	6.99 (6.74–7.24)	6.96	572
33b	Longitudinal Valley fault (Central)	R/SS	1789.01	7.26 (7.01–7.51)	7.31	956
33c	Longitudinal Valley fault (South)	R/SS	806.48	6.95 (6.7–7.2)	6.91	560
33ab	Longitudinal Valley fault (North+Central)	R/SS	2702.54	7.42 (7.17–7.67)	7.55	961
33bc	Longitudinal Valley fault (Central+South)	R/SS	2595.49	7.4 (7.15–7.65)	7.52	950

ID [#1]	***Seismogenic structure name***	***Type*** [#2]	***Area (km^2)***	***M_w(W&C)***	***M_w(Y&M)***	***Recurrence interval (yr)*** [#3]
33	Longitudinal Valley fault (north, central, south)	R/SS	3509.02	7.52 (7.27–7.77)	7.7	990
43+45[#4]	Youchang structure, Fengshan structure	R/SS, SS/R	487.69	6.72 (6.49–6.95)	6.69	932
O54A+O54B	Northern Binhai fault, Southern Binhai fault	RL/R, RL/R	10909.74	7.96 (7.71–8.21)	8.32	7900

Note: #1: IDs beginning with "O" denote offshore seismogenic structures.

#2: RL/R means a reverse fault with a right-lateral component.

#3: Used Mw(Y&M) and mean geologic slip rate.

#3: Structures have geodetic slip rates; however, only the results for geologic slip rates are presented in this table.

Table 3. Year of occurrence (Last event) and associated seismogenic structure for earthquake events used in the BPT model, with examples of mean M_W (Y&M) and mean long-term slip rate. Compared with the Poisson process over the next 50 years, the table also presents the probability change (Recurrence rate change) after accounting for the time-dependent BPT model.

ID	**Time elapsed (yr)**	**M_w (Y&M)**	**Recurrence interval (yr)**	**Poisson**	**Time-dependent**	**Recurrence rate change**	**Multiple[#2]**	**Recurrence interval (yr) after multiple**	**New recurrence rate**
13 +	90	6.8	746	0.064805	0.000162	-99.76%	**13**+14, **13**+15	4785	5.05E-07
15		6.83	2298	0.021515	3.29E-14	-100.00%	10+**15**, 13+**15**		3.16E-16
01	331	7.02	807	0.06007	0.043224	-28.67%	**01**+O01	1715	0.000416
01[#1]			861	0.056444	0.03457	-39.45%		1828	0.000331
16	176	7.77	2652	0.018673	3.77E-10	-100.00%	-		7.52E-12

16#1			1351	0.036324	4.12E-05	-99.89%			8.26E-07
17	25	7.81	727	0.066393	2.77E-08	-100.00%	-		5.52E-10
17#1			1859	0.026541	2.34E-21	-100.00%			2.22E-18
20	119	6.4	258	0.1756	0.197821	14.15%	**20**+21	982	0.001161
20#1			480	0.098955	0.02102	-79.61%		1821	0.000112
22	163	6.8	180	0.241528	0.42665	101.22%	19+**22**	440	0.004566
24	78	6.22	200	0.221199	0.219175	-1.04%	**24**+25	436	0.002273
24#1			241	0.187427	0.131266	-32.20%		525	0.001292
32	73	6.53	75	0.482683	0.759806	116.40%	-		0.028571
	7				0.385151	-26.21%	-		0.009709
33	73	7.7	369	0.126677	0.01671	-87.56%	**33a,**	572	0.000218
							33b,	956	0.000130

							33c,	560	0.000222
							33a+33b,	961	0.000130
							33b+33c,	950	0.000131
							33a+33b+33c	990	0.000126

Note: #1: Used mean geodetic slip rate.

#2: Indicates with which structures (shown in bold by their IDs) this structure can participate in multiple-structure ruptures.

Table 4. Parameters for offshore seismogenic structures.

ID	*Seismogenic structure name*	*Type* [#1]	*Length (km)*	*Width (km)*	*Area (km^2)*	*M_w (W&C)*	*M_w (Y&M)*	*Slip rate (mm/yr)*	*Recurrence interval (yr)* [#2]
1	Outer Chinshan structure 1 (ST-I)	N	23.08	11.32	261.17	7.01 (±0.25)	6.42	0.82 (0.42–2.45)	821
2	Outer Chinshan structure 2	N	120.00	11.32	1357.93	6.40 (±0.25)	7.15	0.64 (0.33–2.78)	2506
3	Outer Kanchiao structure (ST-II)	N	35.20	11.43	402.39	7.13 (±0.25)	6.60	0.36 (0.16–1.47)	2326
4	Outer Keelung structure 1	N	50.70	11.32	573.72	6.59 (±0.25)	6.76	0.39 (0.16–1.96)	2538
5	Outer Keelung structure 2	N	102.81	10.74	1104.05	6.74 (±0.25)	7.04	0.71 (0.31–2.9)	1934
6	Outer Shenaokeng structure 1	N	40.01	11.32	452.70	7.03 (±0.25)	6.66	0.4 (0.17–1.96)	2193
7	Outer Shenaokeng structure 2	N	98.20	10.74	1054.54	6.64 (±0.25)	7.02	0.4 (0.18–1.57)	3390
8	Outer Longdong structure 1	N	31.75	11.32	359.29	7.01 (±0.25)	6.56	0.29 (0.12–1.47)	2681
9	Outer Longdong structure 2	N	45.23	10.74	485.71	6.54 (±0.25)	6.69	0.22 (0.11–0.63)	4132

ID	Seismogenic structure name	Type [#1]	Length (km)	Width (km)	Area (km^2)	M_w (W&C)	M_w (Y&M)	Slip rate (mm/yr)	Recurrence interval (yr) [#2]
10	Outer Aoti structure 1	N	80.68	10.62	857.08	6.67 (±0.25)	6.93	0.76 (0.3–5.38)	1590
11	Outer Aoti structure 2	N	58.08	10.16	590.19	6.92 (±0.25)	6.77	0.18 (0.09–0.44)	5747
12	North Yilan Ridge structure A	N	58.71	16.40	962.62	6.76 (±0.25)	6.98	0.93 (0.39–4.31)	1391
13	North Yilan Ridge structure B	N	28.00	15.93	446.20	6.97 (±0.25)	6.65	0.33 (0.14–1.57)	2639
14	North Yonaguni Rift structure	N	62.04	13.05	809.51	6.63 (±0.25)	6.91	1.85 (1.06–3.13)	639
15	South Yonaguni Rift structure	N	73.79	13.05	962.82	6.90 (±0.25)	6.98	1.85 (1.06–3.13)	698
16	East Yonaguni Rift structure A	N	30.78	15.01	462.04	6.97 (±0.25)	6.66	0.87 (0.43–1.44)	1028
17	East Yonaguni Rift structure B	N	64.90	15.01	974.22	6.65 (±0.25)	6.99	0.87 (0.43–1.44)	1490
18	Suao fault zone	RL	49.30	22.08	1088.74	6.98 (±0.23)	7.04	13.5 (3.4–23.6)	101
19	West Yaeyama fault zone	RL	93.88	14.15	1328.76	7.08 (±0.23)	7.14	19.41 (5.26–33.57)	81

ID	Seismogenic structure name	Type [#1]	Length (km)	Width (km)	Area (km²)	M_w (W&C)	M_w (Y&M)	Slip rate (mm/yr)	Recurrence interval (yr) [#2]
20	Yaeyama fault zone	RL	152.13	13.45	2046.33	7.17 (±0.23)	7.39	16.9 (4.84–28.95)	144
21	Ryukyu Upper Plate splay structure	R	161.30	16.00	2580.80	7.36 (±0.25)	7.52	26.54 (7.74–45.35)	115
22b	Takangkou High thrust	R	95.75	11.63	1113.16	7.40 (±0.25)	7.05	0.35 (0.3–0.39)	4000
23	Chimei Canyon thrust	R	115.76	37.62	4355.12	7.07 (±0.25)	7.82	19 (17.27–21.11)	272
24	East Taitung structure	R(LL?)	53.34	17.68	942.89	7.61 (±0.25)	6.97	0.08 (0.05–0.16)	15152
25	East Lutao structure	R	41.57	42.60	1770.80	7.01 (±0.25)	7.30	4.8 (2.09–8.52)	437
26	East Lanyu structure	R	129.50	42.60	5516.70	7.25 (±0.25)	7.96	5.2 (3.49–7.57)	1258
27	East Huatung Ridge structure	R	193.50	22.00	4257.07	7.70 (±0.25)	7.81	1.59 (1.08–2.29)	3185
28	East Southern Longitudinal Trough structure	R	29.29	17.40	509.62	7.60 (±0.25)	6.71	1.45 (0.61–2.29)	648

ID	Seismogenic structure name	Type [#1]	Length (km)	Width (km)	Area (km^2)	M_w (W&C)	M_w (Y&M)	Slip rate (mm/yr)	Recurrence interval (yr) [#2]
29	Southwest offshore structure A	R	88.20	25.00	2205.00	6.77 (±0.25)	7.43	2.33 (0.92–3.73)	1122
30	Southwest offshore structure B-1	R	48.00	22.00	1056.00	7.34 (±0.25)	7.02	2.44 (0.95–3.93)	553
31	Southwest offshore structure B-2	R	27.70	20.00	554.00	7.05 (±0.25)	6.74	3.71 (1.52–5.9)	264
32	Southwest offshore structure B-3	R	20.40	23.00	469.20	6.80 (±0.25)	6.67	2.79 (1.08–4.51)	322
33	Southwest offshore structure C	R	121.90	22.00	2681.80	6.73 (±0.25)	7.54	2.95 (1.2–4.71)	1076
34	Southwest offshore structure D	R	28.80	19.00	547.20	7.42 (±0.25)	6.74	1.63 (0.68–2.58)	595
35	Southwest offshore structure E	R	131.30	21.20	2784.10	6.79 (±0.25)	7.56	1.03 (0.45–1.62)	3185
36	Southwest offshore structure F-1	R	52.70	22.60	1190.96	7.43 (±0.25)	7.08	1.07 (0.43–1.71)	1340
37	Southwest offshore structure F-2	R	73.49	20.08	1475.37	7.10 (±0.25)	7.20	2.04 (0.89–3.19)	858
38	Southwest offshore structure F-3	R	76.00	20.95	1592.02	7.18 (±0.25)	7.24	2.94 (1.32–4.56)	643

ID	Seismogenic structure name	Type [#1]	Length (km)	Width (km)	Area (km^2)	M_w (W&C)	M_w (Y&M)	Slip rate (mm/yr)	Recurrence interval (yr) [#2]
39	Southwest offshore structure G	R	69.87	23.30	1627.71	7.21 (±0.25)	7.25	1.77 (0.8–3.03)	1089
40	Southwest offshore structure H-1	R	57.86	23.30	1347.92	7.22 (±0.25)	7.14	1.49 (0.59–2.66)	1072
41	Southwest offshore structure H-2	R	53.50	21.76	1164.00	7.15 (±0.25)	7.07	1.34 (0.56–2.13)	1058
42	Southwest offshore structure H-3	R	28.90	24.22	700.06	7.09 (±0.25)	6.85	1.64 (1.08–2.2)	670
43	Southwest offshore structure H-4	R	24.00	19.45	466.70	6.89 (±0.25)	6.67	2.63 (1.42–3.84)	343
44	Southwest offshore structure I	R	67.77	25.22	1709.41	6.73 (±0.25)	7.28	1.63 (0.76–2.7)	1248
45	Southwest offshore structure J	R	24.00	25.22	605.37	7.24 (±0.25)	6.78	1.42 (0.94–1.89)	720
46	Southwest offshore structure K	R	25.00	21.00	525.04	6.83 (±0.25)	6.72	1.63 (0.76–2.7)	584
47	Southwest offshore structure L (East Liuchiu Island structure)	R	41.80	11.27	471.19	6.78 (±0.25)	6.67	1.04 (0.53–1.7)	866

ID	*Seismogenic structure name*	*Type* [#1]	*Length (km)*	*Width (km)*	*Area (km²)*	*M_w (W&C)*	*M_w (Y&M)*	*Slip rate (mm/yr)*	*Recurrence interval (yr)* [#2]
48	Southwest offshore structure M	R	28.20	24.00	676.84	6.74 (±0.25)	6.83	1.38 (0.57–2.4)	780
49	Fangliao Canyan structure Zone	LL/R	63.88	9.14	583.77	6.88 (±0.23)	6.77	0.7 (0.4–2.26)	1429
50	Manila Upper Plate splay structure	R	284.80	23.34	6646.03	6.80 (±0.25)	8.07	0.51 (0.3–0.81)	15625
51	Tainan Basin structure	N	53.40	20.49	1094.25	7.77 (±0.25)	7.04	0.85 (0.01–1.91)	1616
52	Outer Hsinchu structure	R	13.50	14.14	190.92	7.03 (±0.25)	6.28	0.4 (0.18–1.57)	1425
53	Toufen structure	R	20.14	9.33	187.99	6.38 (±0.25)	6.27	1.38 (0.59–6.64)	413
54A	Northern Binhai fault zone	RL/R	236.90	23.86	5651.69	6.38 (±0.23)	7.97	3.19 (2.15–5.17)	2101
54B	Southern Binhai fault zone	RL/R	220.40	23.86	5258.05	7.81 (±0.23)	7.93	1.54 (1.18–2.21)	4049

820 *Note: #1: LL, a left-lateral strike-slip fault; LL/R, a reverse fault with a left-lateral component; and RL, a right-lateral strike slip fault; RL/R, a reverse*

821 *fault with a right-lateral component.*

#2: Used Mw(Y&M) and mean slip rate

Table 5. Parameters of Taiwan subduction zone interface sources. Although this table presents the mean $M_W\_Strasser(A)$ and mean slip rate as example, the analysis actually uses three types of M_W values (including $\pm 1\sigma$) and three values of slip rate for each subduction zone: Ryukyu slip rates are 30, 25, and 15 mm/yr; Manila slip rates are 24, 14, and 8 mm/yr.

<table>
<tr><th colspan="2">ID</th><th>Area (km^2)</th><th>M_W</th><th>Slip Rate (mm/yr)</th><th>Recurrence Interval (yr)</th></tr>
<tr><td rowspan="6">Ryukyu</td><td>T01</td><td>16974</td><td>8.27</td><td rowspan="6">25</td><td>1178</td></tr>
<tr><td>T02</td><td>41497</td><td>8.6</td><td>1599</td></tr>
<tr><td>T03</td><td>60027</td><td>8.73</td><td>1228</td></tr>
<tr><td>T01, T02</td><td>58471</td><td>8.72</td><td>620</td></tr>
<tr><td>T02, T03</td><td>101524</td><td>8.93</td><td>710</td></tr>
<tr><td>T01, T02, T03</td><td>118497</td><td>8.98</td><td>1175</td></tr>
<tr><td rowspan="6">Manila</td><td>T04</td><td>27918</td><td>8.45</td><td rowspan="6">14</td><td>1178</td></tr>
<tr><td>T05</td><td>20112</td><td>8.33</td><td>1599</td></tr>
<tr><td>T06</td><td>25376</td><td>8.42</td><td>1228</td></tr>
<tr><td>T04, T05</td><td>48030</td><td>8.65</td><td>620</td></tr>
<tr><td>T05, T06</td><td>45489</td><td>8.63</td><td>710</td></tr>
<tr><td>T04, T05, T06</td><td>73406</td><td>8.81</td><td>1175</td></tr>
</table>

Table 6 The upper and lower depth limits of each subregion in subduction intraslab.

ID [#1]	**Upper depth (km)**	**Lower depth (km)**
NP1	35	63
NP2	35	95
NP3	63	165
NP4	110	165
NP5	165	300
SP1	40	80
SP2	40	130
SP3	65	190
SP4	130	220

Note: #1: IDs beginning with "NP" denote Ryukyu subduction intraslabs; beginning with "SP" denote Manila subduction intraslabs.

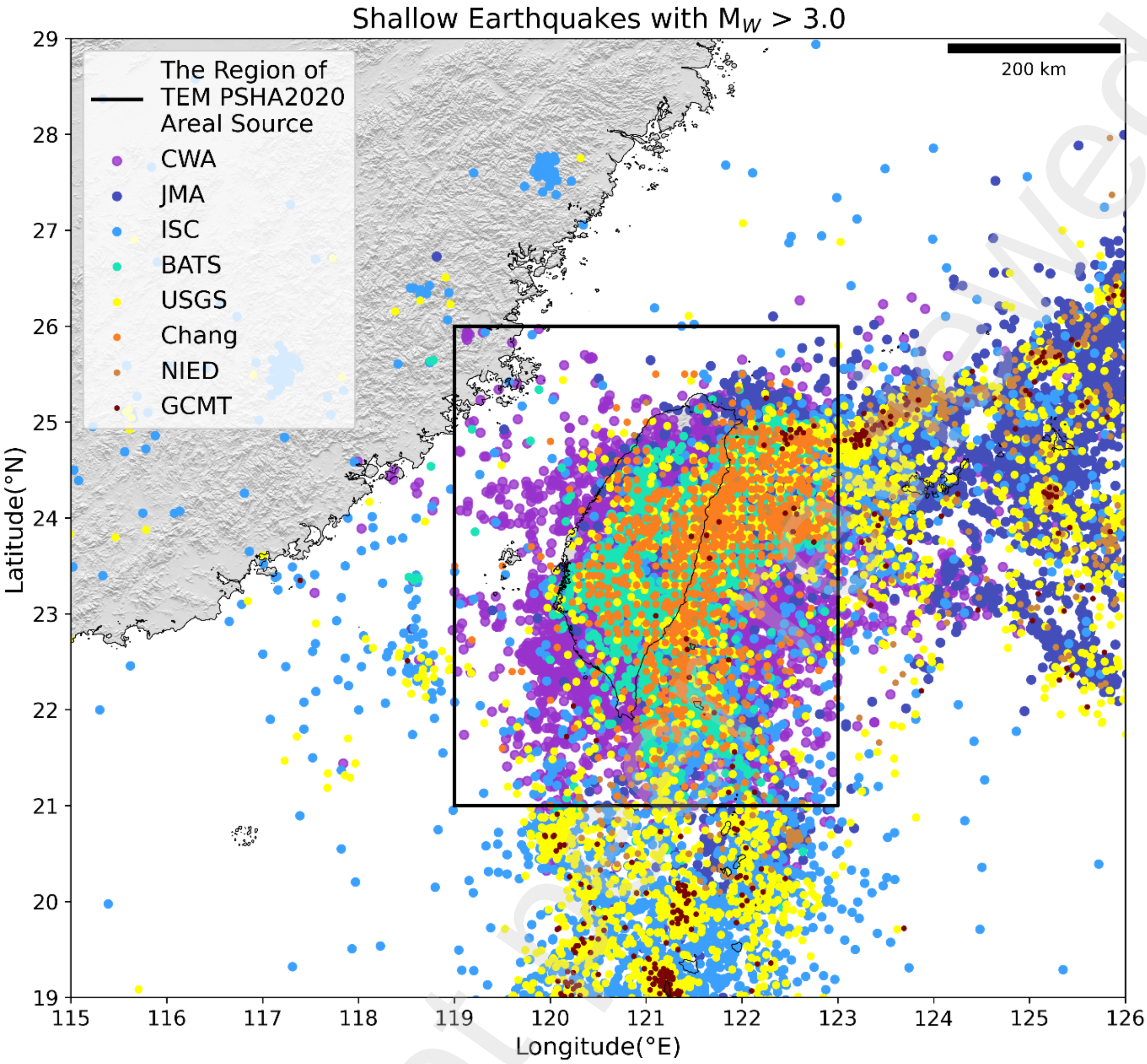


Fig. 1. Study area of shallow-background sources in the TEM PSHA2020 (black box) and the spatial distribution of merged catalog with M_W greater than 3.0 and depths less than 35 km from various network.

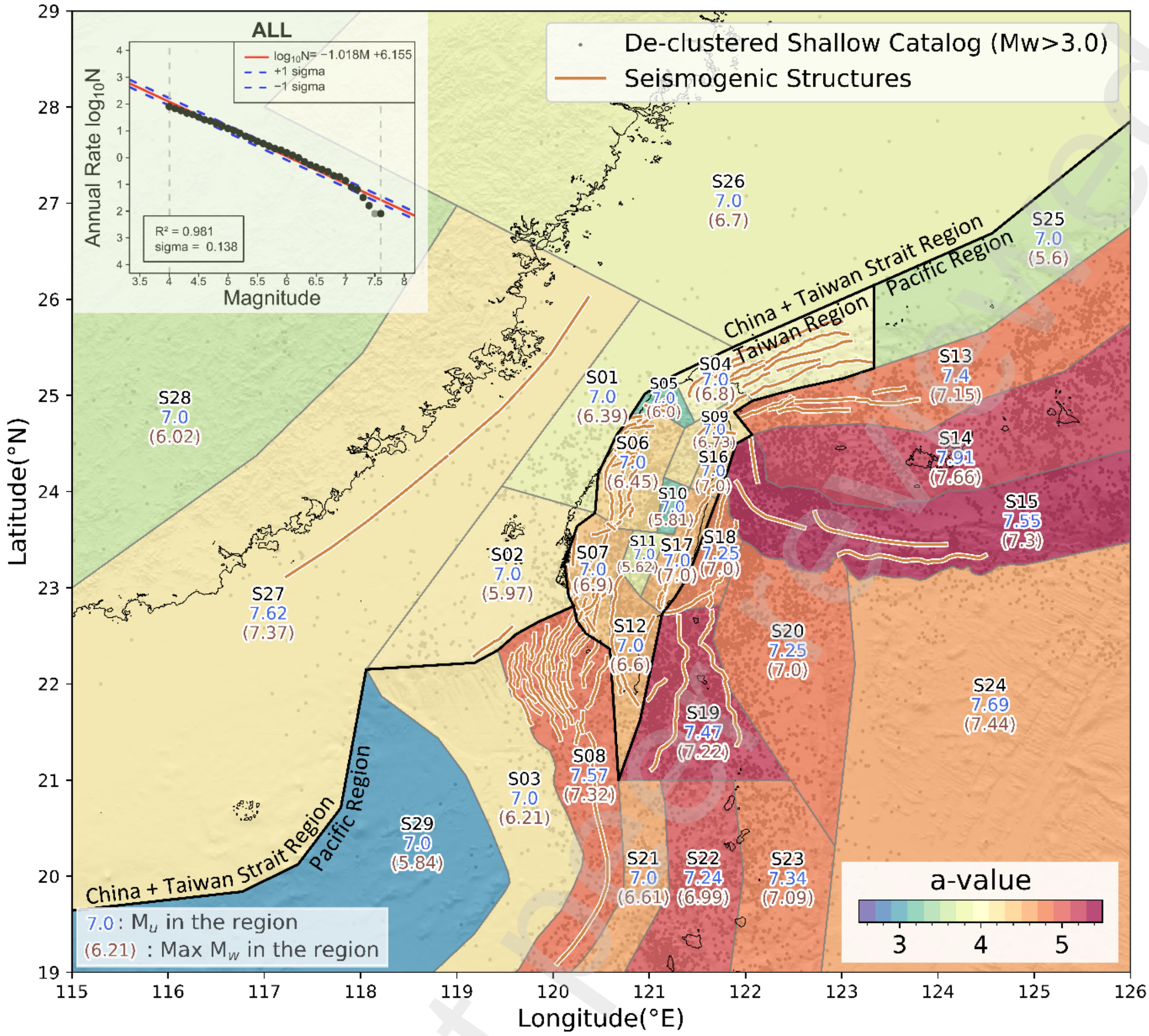


Fig. 2. The figure shows a-values, upper magnitude limits (M_u), and the maximum recorded magnitude (Max M_W) from 1900 to 2023 for each subregion of the areal source. The overall b-value for the study area is 1.018; the Gutenberg-Richter relation is shown in the upper left corner. Earthquake distribution (black dots), fault traces (brown lines), and boundaries (black lines) of the three major regions used to analyze the magnitude completeness are also illustrated.

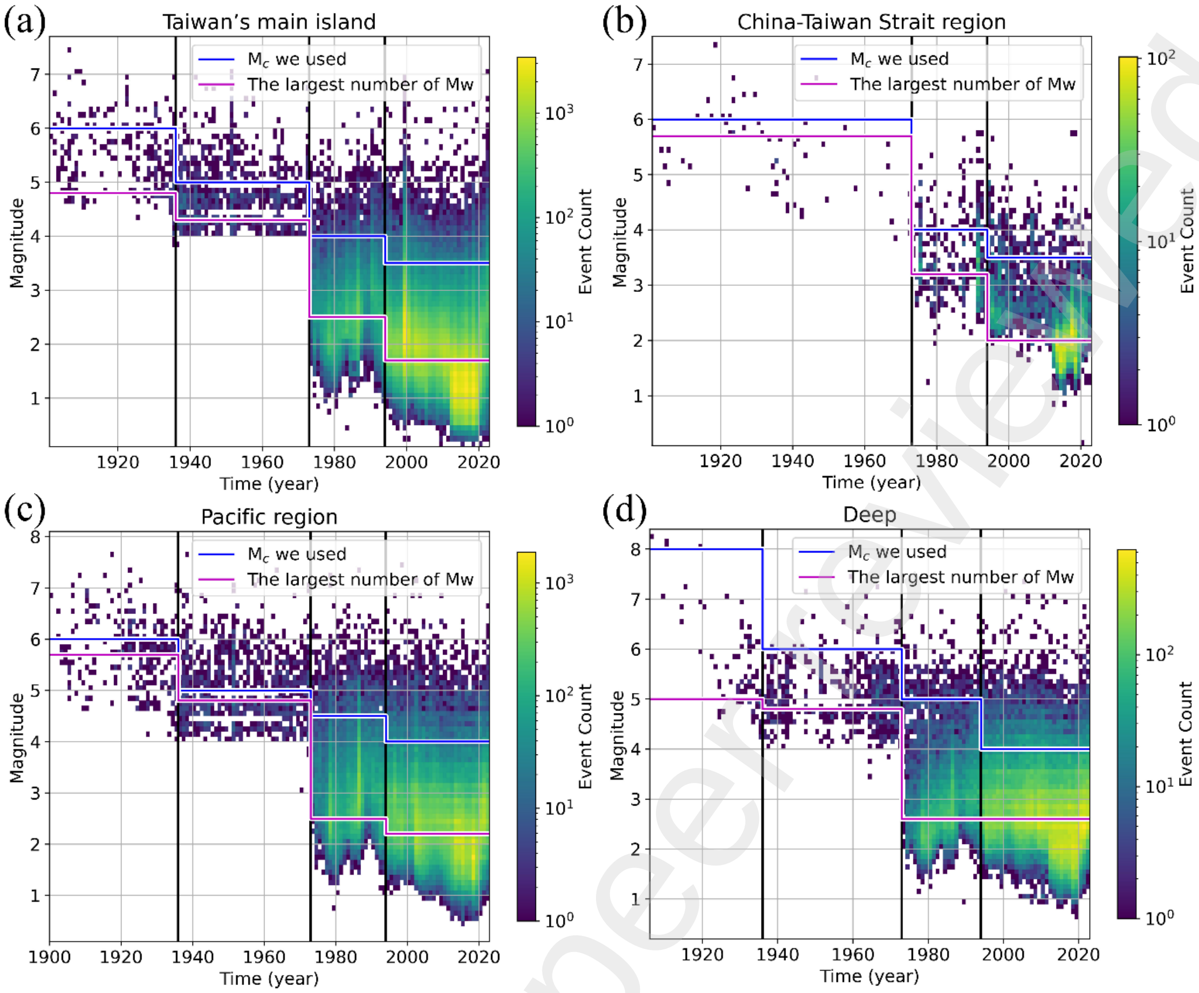


Fig. 3 M_C for different time periods of shallow earthquakes, with shallow events further divided into three main regions (a, b, c); (d) M_C for different time periods of deep earthquakes.

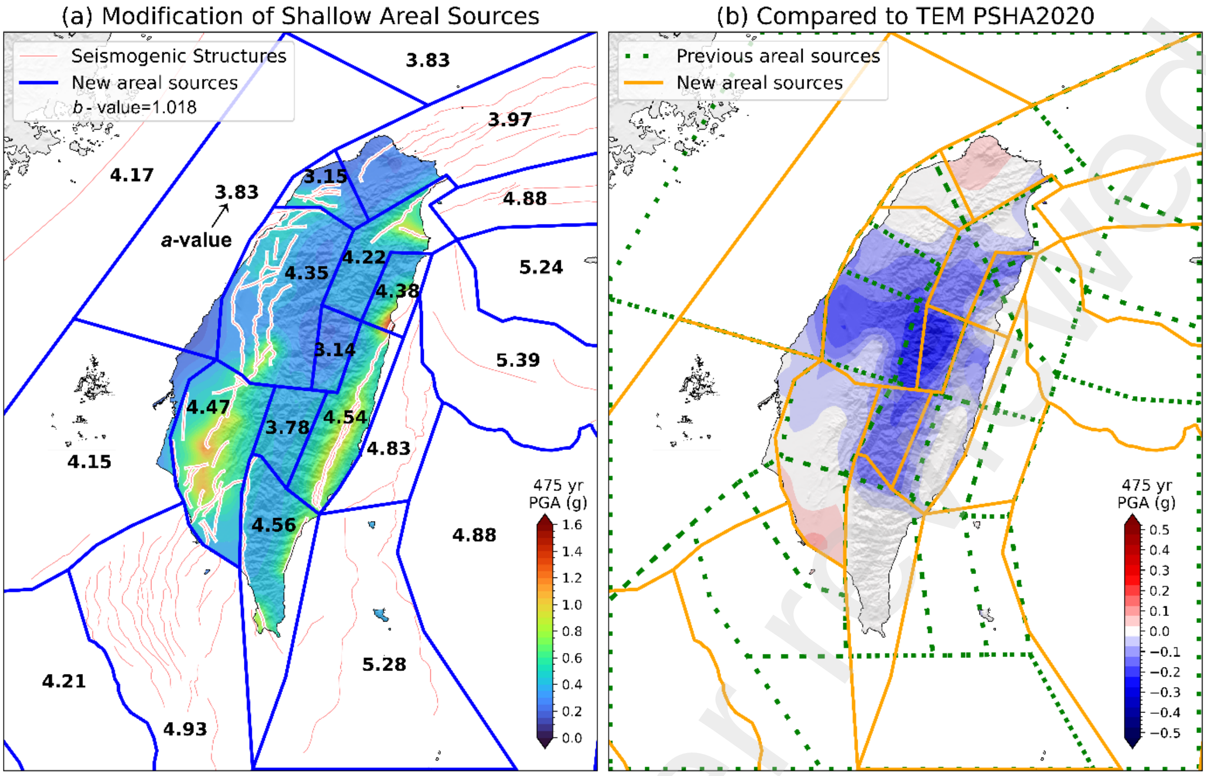


Fig. 4 The seismic hazard map (a) considering the new areal sources, including newly defined areal sources around the main island of Taiwan (blue lines), new seismogenic structure traces (pink lines), and (b) the impact of these updates; areas shown in blue represent regions where hazard has decreased following the updates.

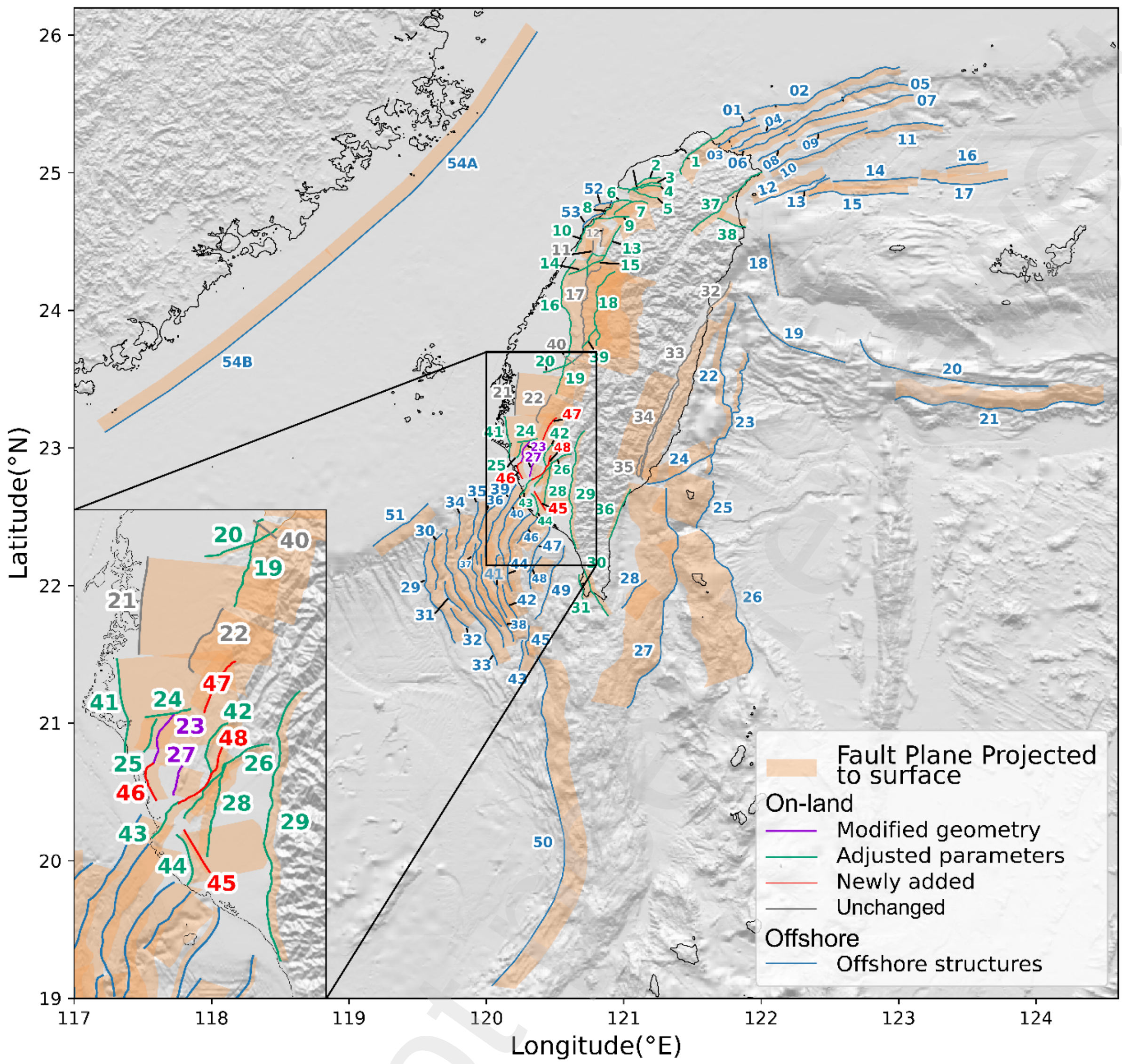


Fig. 5 New fault distributions, surface projections of fault planes, and fault numbers are shown. Blue lines indicate offshore seismogenic structures, while all other color lines are on-land seismogenic structures: dark-violet for faults with modified geometry and parameters; seafoam green for faults with parameter adjustments; red for newly added faults; gray for unchanged faults

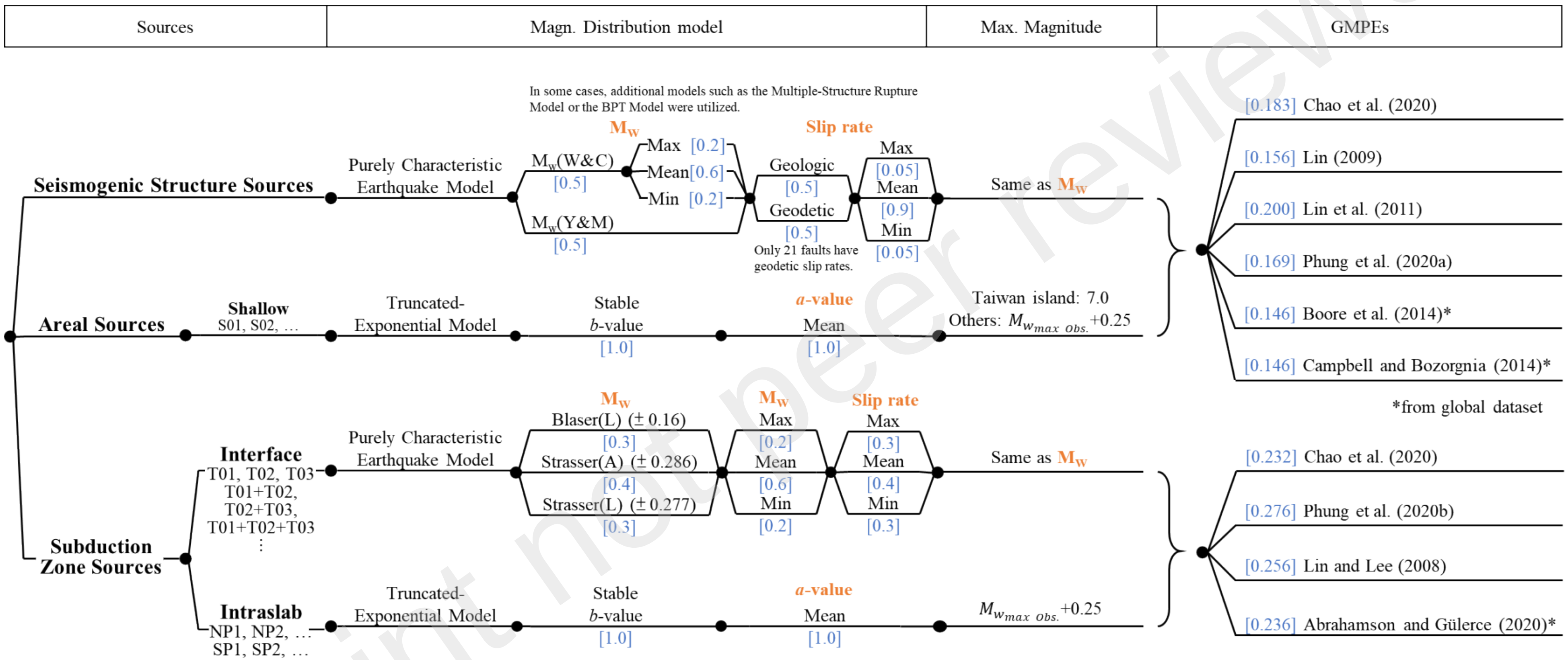


Fig. 6. The complete logic tree of TEM PSHA2025. Numbers in blue brackets indicate the weights of the logic tree branches.

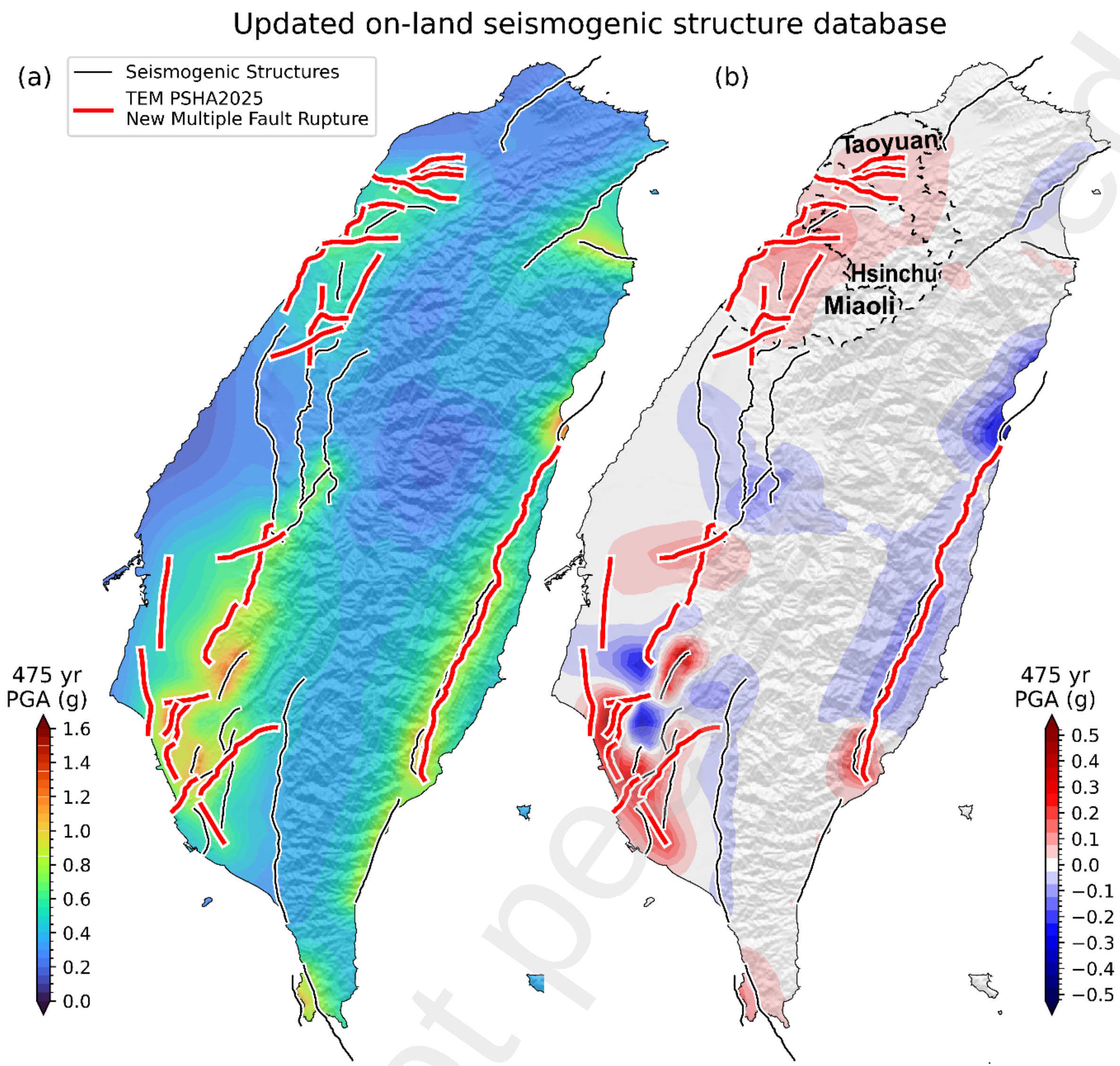


Fig. 7 The seismic hazard map (a) considering the new on-land seismogenic structures, including newly defined pairs of multiple-structure ruptures (red lines), used the Fig. 6 logic tree to account uncertainties but without geodetic slip rate, and (b) the impact of these updates.

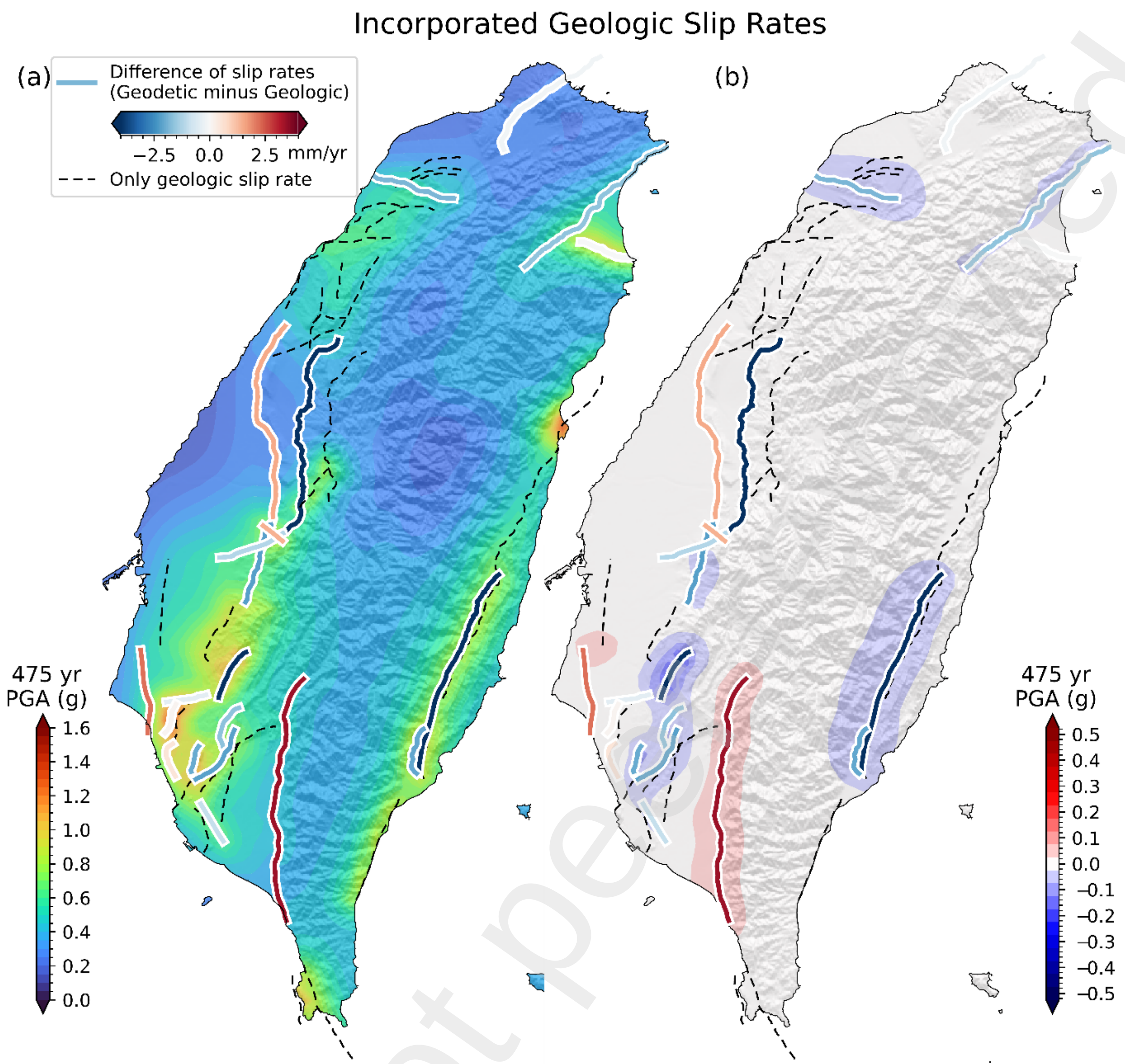


Fig. 8 The seismic hazard map considering new on-land seismogenic structures is shown in (a), where 21 faults (colored lines) are assigned weights of 0.5 each for geologic and geodetic slip rates (see Fig. 6). The colors of the lines represent the differences between geologic and geodetic slip rates for these faults: blue indicates geodetic slip rates are lower than geologic, red indicates the reverse, and lines closer to white represent smaller differences. Figure (b) illustrates the impact of this innovation.

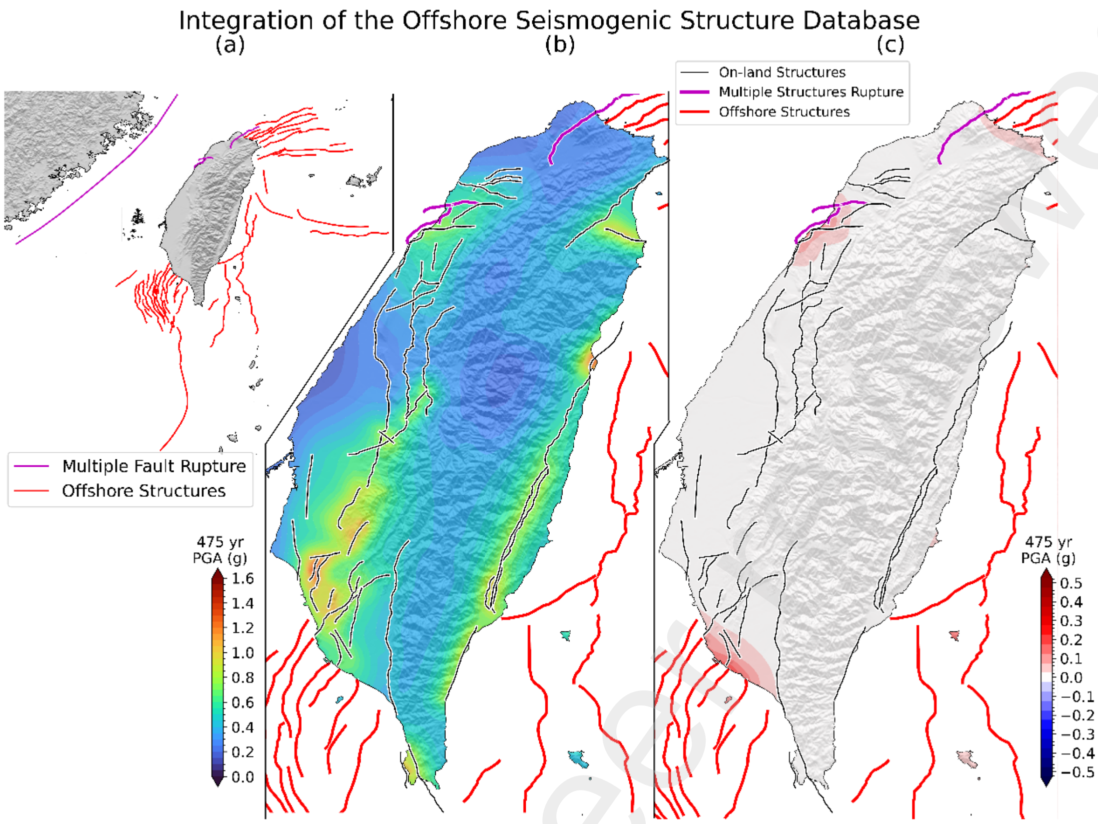


Fig. 9 (a) Complete distribution of offshore seismogenic structures and multiple-structure ruptures (blue lines), together with (b) the updated seismic hazard map incorporating newly identified offshore seismogenic structures (red lines). The logic-tree treatment for offshore faults follows the same framework as for onshore seismogenic structures, except that geodetic slip rates are not included (cf. Fig. 6). (c) illustrates the impact of incorporating offshore seismogenic structures into the hazard assessment.

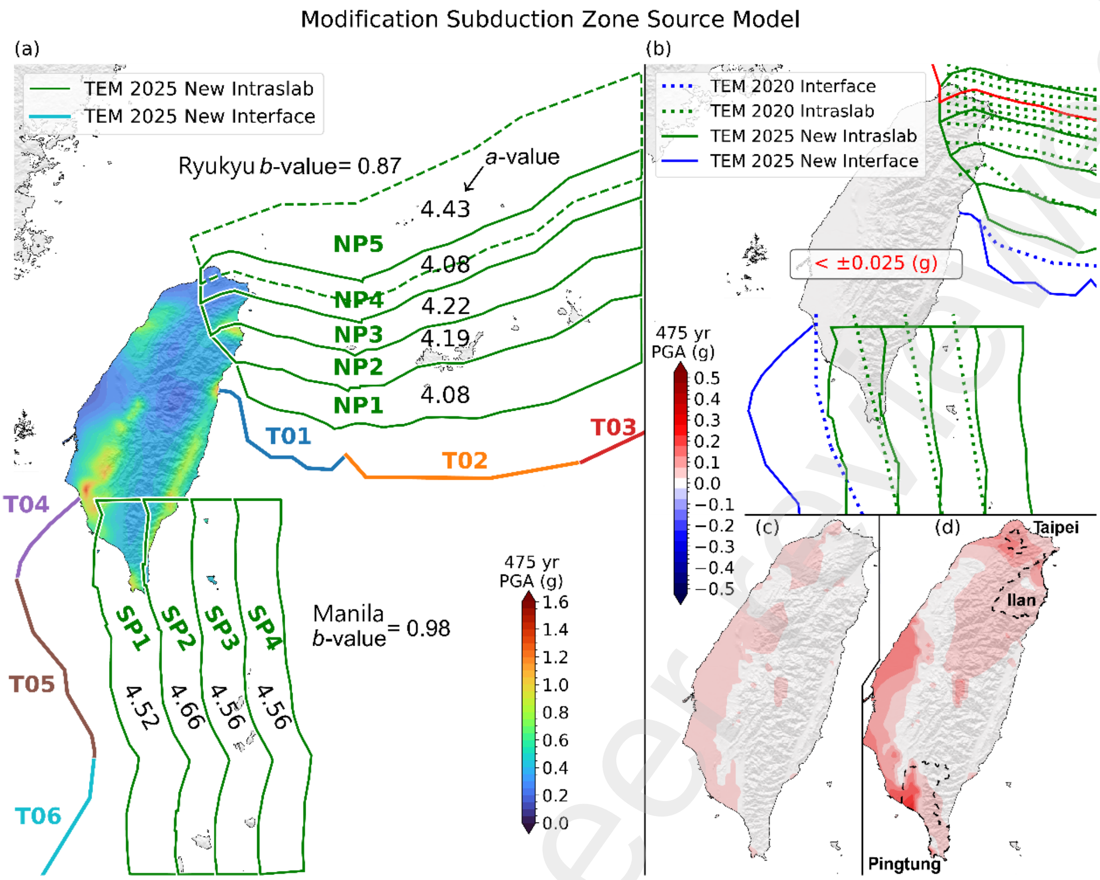
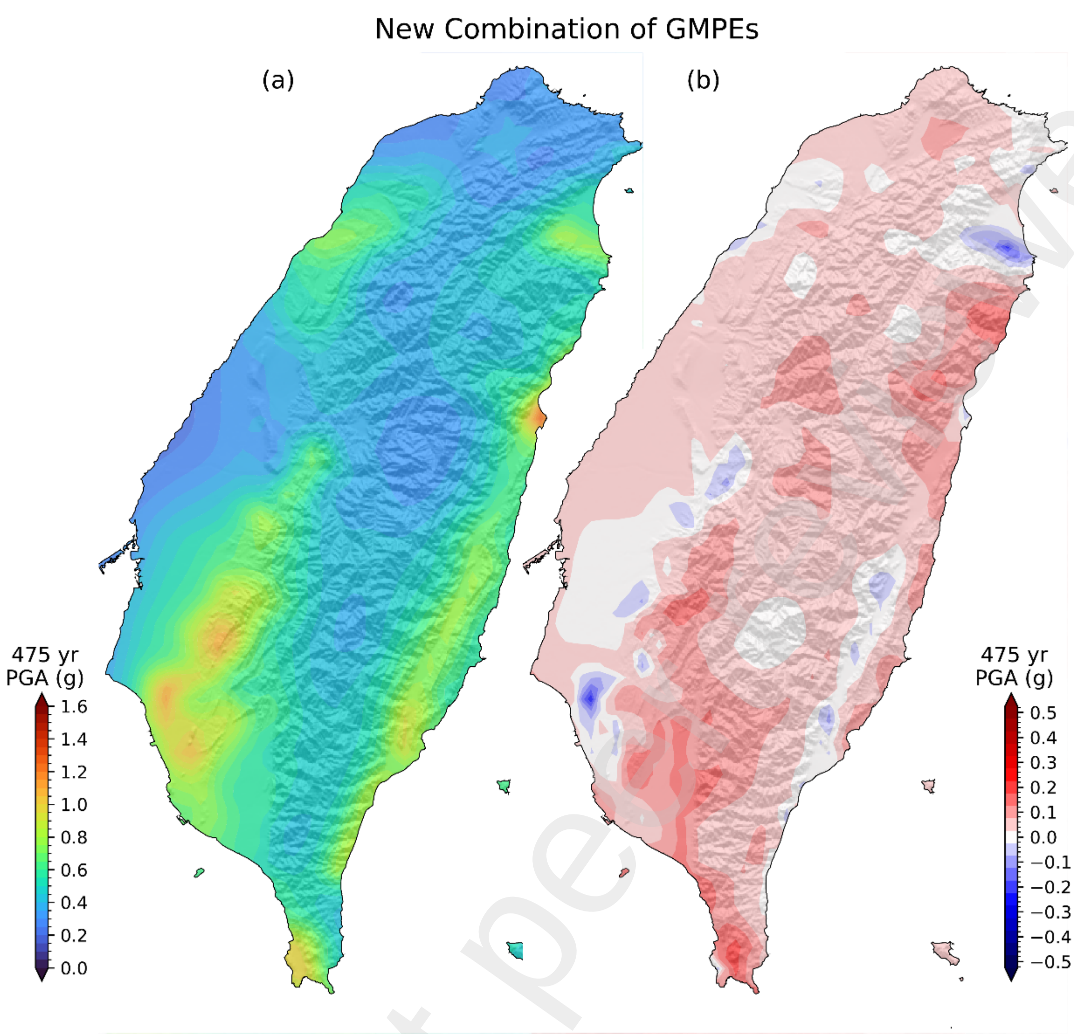


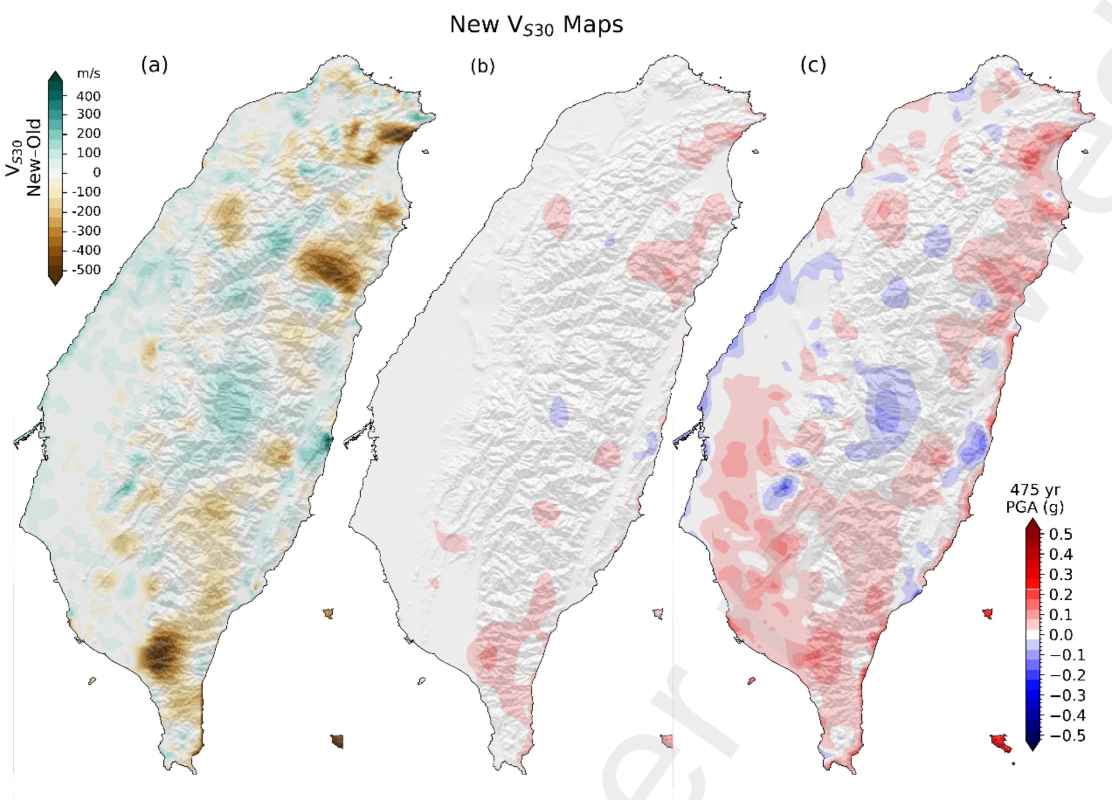
Fig. 10 The seismic hazard map (a) considering new subduction zone sources, including interface sources (T01–T06) and intraslab sources (NP1–5, SP1–4), with *a*- and *b*-values for the intraslab sources; (b) shows the impact of this innovation in PGA for a 10% probability in 50 years. All the differences are within 0.025g, indicating negligible change, but (c) shows greater differences for SA1.0, and when analyzing (d) the 2% probability in 50 years, the variations become more pronounced.



899

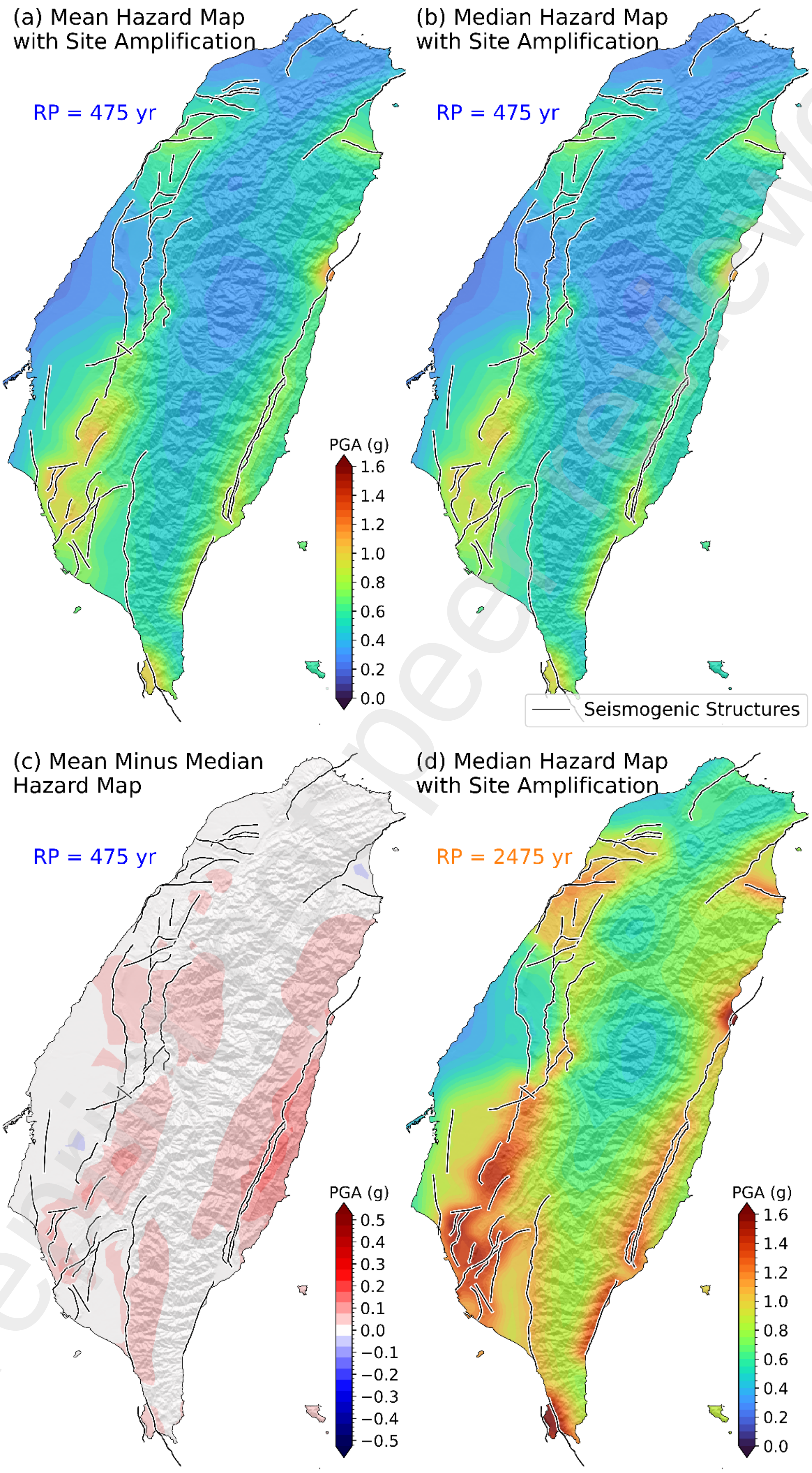
900 Fig. 11 The seismic hazard map (a) considering the new GMPEs and (b) the impact of
901 this innovation. The GMPEs are selected and weighted according to the method of
902 Gao et al. (submitted).

903

904 

905 Fig. 12 (a) Differences between the new and previous (TEM PSHA2020) V_{S30} maps;
906 (b) the impact on the seismic hazard map for PGA, and (c) for SA1.0 using the new
907 V_{S30} map at 10% probability in 50 years.

908

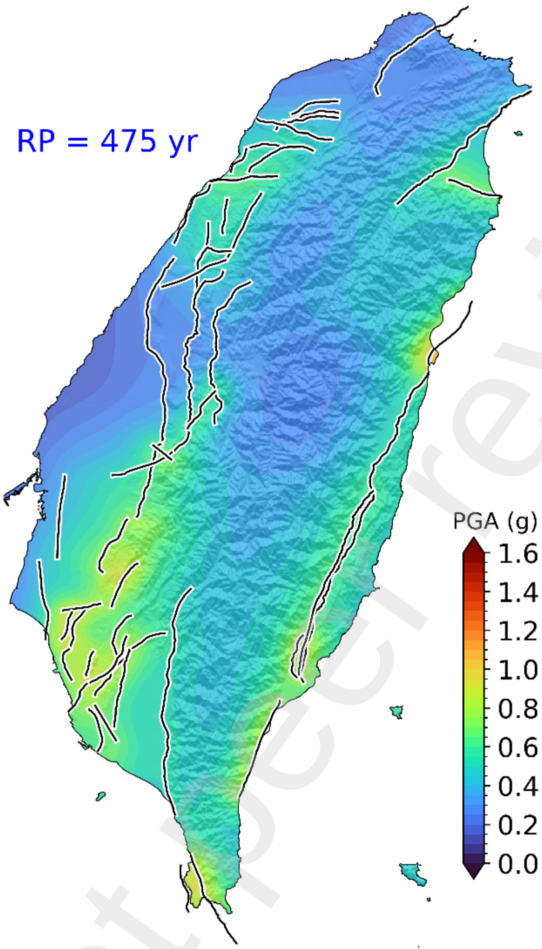


(a) Mean Hazard Map
with Site Amplification
RP = 475 yr
(b) Median Hazard Map
with Site Amplification
RP = 475 yr
PGA (g)
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
Seismogenic Structures
(c) Mean Minus Median
Hazard Map
RP = 475 yr
(d) Median Hazard Map
with Site Amplification
RP = 2475 yr
PGA (g)
0.5
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
−0.4
−0.5
PGA (g)
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0

909

Fig. 13 Final seismic hazard maps from TEM PSHA2025: (a) updated V_{S30} map considering site amplification for a 10% probability in 50 years; (b) median hazard map within the complete TEM PSHA framework; (c) differences between the median and mean values (median minus mean); and (d) median hazard map within the complete TEM PSHA framework for a 2% probability in 50 years. In the figure, RP denotes return period.

917 ## Appendix



918

919 Fig. a Final median seismic hazard maps from TEM PSHA2025 using engineering
920 bedrock (V_{S30} = 760 m/s) for a 10% probability in 50 years. In the figure, RP denotes
921 return period.